

# BEYOND THE BASE: EXPONENT ALLOCATION IN ROPE

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## ABSTRACT

RoPE assigns its rotary pairs frequencies through a base and an exponent sequence. We study how the distribution of those exponents changes positional representation and length generalization. Paired training with identical frequency endpoints shows that changing only the interior exponents improves all three tested extrapolation lengths in all three seeds. Retargeting the range changes the ordering, and frozen weights prefer runtime tables derived from the allocation they trained with. A full sine-cosine subspace analysis characterizes how exponent allocation changes the positional basis, including the shared two-dimensional limit of slow frequencies. We also derive an analytic Cosh family from a specified convex allocation criterion and evaluate this explicit construction in language-model training and adaptation. After matched Llama-3-8B adaptation at 8K, 32K perplexity falls from 991.5 to 127.9. For frozen models, we study exponent displacements relative to the native table. A boundary-matched adjustment improves OLMo’s mean token F1 over MrRoPE-Pro from 21.62% to 25.44% across five natural-QA tasks. The results connect exponent shape, sampled range, and learned compatibility as joint considerations in RoPE design.

## 1 INTRODUCTION

Rotary position embedding (RoPE) assigns a different frequency to each pair of query and key coordinates (Su et al., 2024). Its usual form,  $\omega_k = b^{-\phi_k}$ , combines a base  $b$  with equally spaced exponents. Base-selection and context-scaling methods have established that the range of positional scales matters (Liu et al., 2024; Xu et al., 2024; Oka et al., 2026b). A finite table also makes a distributional choice: how its rotary pairs occupy that range. We study this choice through the exponent sequence.

Three questions organize the study. What changes in the positional basis when exponents are redistributed? Does this redistribution affect learned behavior when the frequency range is fixed? And how does an already trained model respond to adjustments of its native exponents? These questions connect geometric analysis, controlled interventions, and practical table design.

Figure 1 shows the basic intervention. Two allocations share their endpoints and 32 rotary pairs; only the 30 interior exponents differ. Across three paired training seeds, this change lowers NLL at each tested extrapolation length. A separate range-retargeted evaluation reverses the ordering. The weights-by-table crossings add another finding: the preferred runtime table depends strongly on which allocation trained the weights. Together, these controls make exponent shape observable while revealing its interaction with range and learned representation.

The positional geometry provides a way to analyze these distributions. Each rotary pair supplies an entire sine-cosine subspace. Canonical correlations compare these subspaces without privileging a content phase, and expose the common low-frequency limit  $\text{span}\{1, \Delta\}$ . Thus many slow pairs can supply few distinct positional directions over a bounded interval. The learned content coefficients determine how the model uses those directions.

We then study concrete exponent designs in two settings. A convex cumulative-tail criterion yields an analytic Cosh distribution, whose quantiles define a fixed table for training and adaptation. Its experiments cover scarce rotary channels, continued training, and matched 8B adaptation. For frozen models, we instead construct displacements from the native exponents and compare their placement and intermediate-band shape. These comparisons include a static Qwen table-amplitude configuration that improves 64K RULER over YaRN while retaining the 32K score, and a boundary-matched

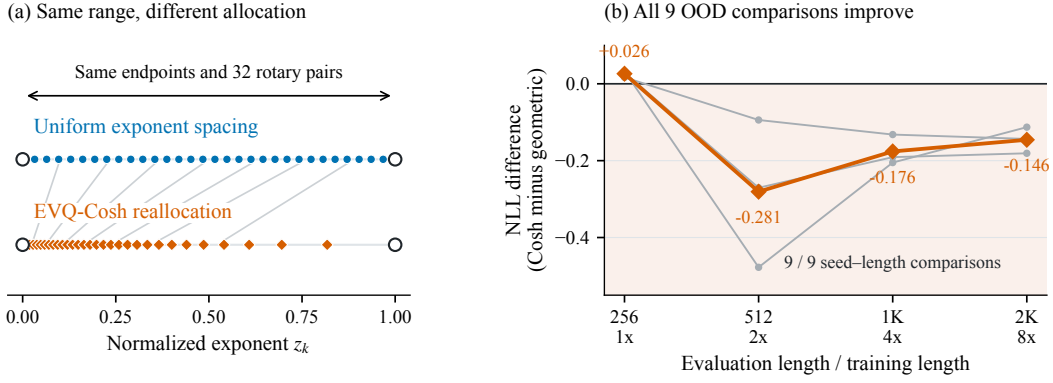


Figure 1: **Interior exponents change learned behavior at fixed frequency range.** Left: two allocations of the same 32 rotary pairs with identical endpoints. Right: paired 151.9M models trained at length 256; changing the 30 interior exponents lowers extrapolation NLL in every seed. Thin lines show the three training seeds and the heavy line their mean.

adjustment with improvements on five OLMo natural-QA tasks. The tested Qwen checkpoints favor a different intermediate profile, making learned compatibility part of the exponent-design problem.

**Contributions.** We (i) define exponent allocation and analyze its full-subspace positional geometry; (ii) identify its effects through fixed-range, cross-shape, and weights-by-table controls, and evaluate a closed-form construction in trained models; and (iii) develop and test native-relative exponent adjustments in frozen models, including a boundary-matched intermediate-band rule.

## 2 EXPONENT ALLOCATION

Let  $K = d_{\text{rot}}/2 \geq 2$  be the number of rotary pairs and  $b > 1$  the base. Standard RoPE uses

$$\omega_k = b^{-\phi_k}, \quad \phi_k = \frac{k}{K}, \quad k = 0, \dots, K-1. \quad (1)$$

The base controls the spacing of this geometric frequency family. Allowing  $\phi_k$  to follow a non-linear sequence also changes the density of samples across positional scales. To describe that shape without conflating it with an overall shift or stretch, take an ordered table  $\omega_0 > \dots > \omega_{K-1} > 0$  and write

$$x_k = -\log \omega_k = a + Rz_k, \quad z_k = \frac{x_k - x_0}{x_{K-1} - x_0}, \quad z_0 = 0, \quad z_{K-1} = 1. \quad (2)$$

Here  $a = x_0$  and  $R = x_{K-1} - x_0$  specify the sampled log-frequency range. Geometric tables have  $z_k = k/(K-1)$  at every base. Holding  $(a, R)$  fixed while changing  $z$  therefore changes only the normalized exponent allocation. The standard exponent grid  $k/K$  in (1) and its endpoint-normalized shape  $k/(K-1)$  thus describe the same geometric table.

This representation gives both a construction and an experimental control. A density over exponents generates a finite table through its quantiles. The same quantiles can either be installed directly or affinely mapped onto fixed endpoints to test the effect of their shape. We use the anchored version for fixed-range comparisons and state the installed grid in each model protocol. For a mature checkpoint, the corresponding variable is the displacement from its original exponents; §6 develops that form.

The analysis below asks what positions these finitely many exponents can represent. The experiments then ask how models learn to use the resulting basis and how existing weights respond to a change in it.

## 3 CONTROLLED EFFECTS OF EXPONENT ALLOCATION

We begin with interventions that distinguish exponent shape, sampled range, and the weights learned under an allocation. The Cosh quantiles used here are constructed in §4.3; endpoint anchoring keeps their range fixed. An exponential deformation supplies a second nonuniform family.

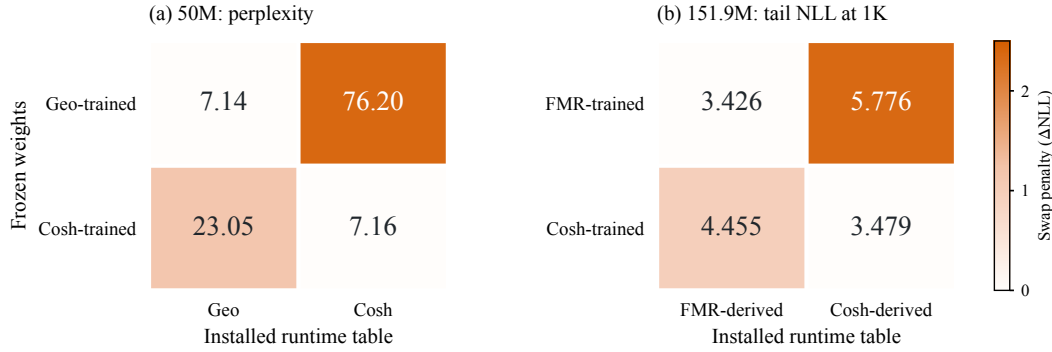


Figure 2: **Weights co-adapt to their exponent allocation.** Each row freezes one trained weight set and changes only its runtime table. Left: the 50M crossing reports PPL on eight windows. Right: the 151.9M crossing reports mean tail NLL on 32 anchors in each of two training seeds, using extensions derived from the two training tables. FMR denotes the geometric table at the fixed training range. Shading shows the NLL increase relative to the matched diagonal cell.

### 3.1 DOES EXPONENT SHAPE MATTER AT FIXED RANGE?

We trained paired 151.9M models at length 256 with  $K = 32$  rotary pairs. Within each of three seeds, architecture, initialization, token order, optimizer, schedule, and the 499,974,144-token budget are identical. The geometric arm follows the FMRoPE training-base rule  $b = L_{\text{train}}$  (Oka et al., 2026b); the Cosh arm anchors its  $\tau = 4$  quantiles to the same two endpoints. Only the 30 interior exponents differ.

At the fixed training range, Cosh lowers tail NLL at 512, 1K, and 2K in all three seeds (Fig. 1). The mean Cosh-minus-geometric differences at 256/512/1024/2048 are  $+0.026/-0.281/-0.176/-0.146$ . Exponent shape therefore changes learned length generalization even when no scalar base change is available to explain the result.

A separate evaluation retargets each table’s range to the declared test length while preserving its normalized exponent shape. This intervention favors the geometric arm, with Cosh-minus-geometric NLL  $+0.060/+0.227/+0.460$  at 512/1K/2K. Range selection and exponent allocation consequently need to be evaluated together when composing a deployment table. The complete paired curves and training details appear in Appendix C.

### 3.2 DIFFERENT ALLOCATION SHAPES ACROSS CONFIGURATIONS

A 50.9M factorial varies two bases, two training lengths, three head dimensions, and three paired seeds, with endpoints pinned in every comparison. It evaluates three Cosh strengths and a deformation-matched exponential. At the reference Cosh strength, the weighted OOD objective improves in 7/12 configurations; the preassigned  $1.25\times$  strength improves 10/12, and the matched exponential improves 9/12. The reference-Cosh-minus-exponential mean difference is  $+0.00074$  NLL, with interval  $[-0.006, 0.008]$  (App. Table 14). The observed means favor several preassigned nonuniform shapes, with configuration-dependent strengths and uncertainty in the cross-configuration contrasts. Table 14 reports the unadjusted paired tests.

### 3.3 WEIGHTS LEARN TO USE THEIR EXPONENT BASIS

A model trained with one exponent distribution has learned query and key coefficients in that basis. The weights-by-table crossing makes this dependence visible: in two 50M models, replacing the Geo-trained model’s table by Cosh changes PPL from 7.14 to 76.20; the reverse swap changes 7.16 to 23.05 (Fig. 2). A 151.9M crossing replicates the preferred-table reversal in two training seeds. These crossings connect exponent design during learning with adjustments of a fixed checkpoint: the installed table acts together with learned coefficients.

## 4 POSITIONAL GEOMETRY OF EXPONENT DISTRIBUTIONS

### 4.1 WHAT DIFFERENT EXPONENTS REPRESENT

A rotary pair contributes  $C \cos(\omega\Delta) + D \sin(\omega\Delta)$  to an attention logit, where  $\Delta$  is the relative position and  $C, D$  depend on query and key content. The positional object associated with its exponent is therefore the subspace

$$V_\omega = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}. \quad (3)$$

Comparing these two-dimensional spaces accounts for every content phase within a pair. Let  $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$ ,  $S_\omega = \mathbb{E}[x_\omega^\top x_\omega]$ , and  $H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu]$  under a specified distribution of separations. For nonsingular  $S_\omega, S_\nu$ , the singular values of  $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$  are the canonical correlations of the two spaces. Their average squared correlation,

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1], \quad (4)$$

is unchanged by rotation of the phase inside either pair. It measures how much of the positional basis one frequency shares with another.

Stacking all block-whitened pairs gives a Gram matrix  $\Gamma$  with  $I_2$  on its diagonal blocks. If  $\bar{c}$  is the mean of  $c_{ij}$  over ordered distinct pairs, its Rényi-2 effective rank satisfies

$$r_2(\Gamma) = \frac{(\text{tr } \Gamma)^2}{\text{tr}(\Gamma^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (5)$$

At fixed  $K$  and range, exponent allocation changes this basis through the off-diagonal correlations. For uniform separations on  $[0, L]$ , the cross-Gram has an exact trigonometric form, so both the correlations and effective rank can be computed directly (App. A.1).

### 4.2 HOW SLOW EXPONENTS SHARE POSITIONAL DIRECTIONS

**Proposition 1** (Shared slow-frequency subspace). *Under the uniform measure on  $[0, L]$ , as  $\omega L \rightarrow 0$  through  $\omega > 0$ , the two-dimensional subspace  $V_\omega$  approaches  $\text{span}\{1, \Delta\}$ . If  $\epsilon = \max\{|\omega L|, |\nu L|\} \rightarrow 0$ , then  $2 - \|Q_{\omega\nu}\|_F^2 = O(\epsilon^4)$ .*

The limit follows from  $\cos(\omega\Delta) \rightarrow 1$  and  $\sin(\omega\Delta)/\omega \rightarrow \Delta$ ; Appendix A.1 derives the leading fourth-order coefficient. Several slow pairs can thus supply many coordinate dimensions but very few distinct positional directions. For  $b = 500,000$ ,  $K = 64$ , and  $L = 4096$ , the 23 standard-grid pairs with  $\omega L \leq 1$  have 46 nominal dimensions and  $r_2 = 2.00013$  under this measure. Block whitening normalizes each pair’s energy and conditioning, so  $r_2$  measures directional overlap; the raw feature scales are analyzed in Appendix A.1.

Exponent redistribution changes how many pairs occupy these shared directions and how many cover faster positional variations. Frequency-use studies also show that slow pairs can carry semantic information (Barbero et al., 2025; Urrutia et al., 2026); their positional similarity motivates redistributing the rotary pairs while retaining their content coordinates. The learned weights determine how the resulting basis is used.

### 4.3 AN ANALYTIC ALLOCATION CONSTRUCTION

To construct a smooth exponent table, let a positive density  $\rho \in C^2([0, 1])$ , with  $\int_0^1 \rho = 1$ , allocate samples over  $[0, 1]$ . Uniform  $\rho$  recovers geometric spacing. Define the cumulative slow-tail mass  $S_\rho(t) = \int_t^1 \rho(\phi) d\phi$  and choose the convex allocation criterion

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho(\phi)^2 d\phi + \frac{\beta}{2} \int_0^1 S_\rho(t)^2 dt, \quad \alpha > 0, \beta \geq 0. \quad (6)$$

The first term discourages concentration in a narrow band; the second penalizes the squared mass beyond each slow-end threshold. Equivalently,  $\int S_\rho^2 = \iint \rho(\phi)\rho(\psi) \min(\phi, \psi) d\phi d\psi$ . These specified design preferences provide a tractable surrogate for constructing a nonuniform allocation.

**Theorem 2** (Cosh allocation). *The unique minimizer of (6) over positive  $\rho \in C^2([0, 1])$  with unit integral is*

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1 - \phi))}{\sinh \tau}, \quad \tau = \sqrt{\beta/\alpha}, \quad (7)$$

with  $\rho_0(\phi) = 1$  obtained by continuity at  $\beta = 0$ .

The Euler–Lagrange equation reduces to  $\rho'' = \tau^2 \rho$  with  $\rho'(1) = 0$ ; normalization fixes its scale. Integrating and inverting the CDF gives the finite table

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \quad u_k = \frac{k + 1/2}{K}. \quad (8)$$

We call this one-parameter inverse-CDF family EVQ-COSH. For  $\tau > 0$ , its density decreases from the fast to the slow end, allocating more samples to faster positional variations. The limit  $\tau \rightarrow 0$  gives uniform-log midpoint quantiles.

After choosing  $\tau$  and a grid, installation evaluates (8) once and sets  $\omega_k = b^{-\phi_k}$ . The frequency table stays fixed during training or adaptation, and the attention operator and trainable parameter count are unchanged. For the fixed-range experiment we normalize these quantiles using (2) and anchor them to the baseline endpoints. Table 7 records each experiment’s  $\tau$ , grid, and selection convention; the operating-rule study is in Appendix A.8.

## 5 MODEL EFFECTS OF AN ANALYTIC ALLOCATION

We apply Cosh quantiles in three settings: an MLA architecture with few rotary channels, continued full-parameter training, and adaptation of a pretrained language model. Together they test how an explicit allocation supports language modeling, retrieval, and source use.

### 5.1 LANGUAGE MODELING AND RETRIEVAL

**Scarce rotary channels.** The 432M MLA model uses 16 rotary pairs and trains on 500M tokens at length 8192 (DeepSeek-AI, 2024a;b). Across three seeds, EVQ-COSH reduces 16K perplexity from 138.8 to 95.6; at 8K the corresponding values are 35.4 and 35.8 (Table 1). This tests exponent allocation in an architecture where only a small set of dimensions carries explicit rotation.

**Continued full-parameter training.** The matched 750M continuation trains both arms at length 4096. At 16K, EVQ-COSH reduces perplexity from 45.1 to 24.4, while the 4K values remain close at 22.0 and 22.3. Across 40 associated 8K passkey trials, greedy autoregressive exact match rises from 0 to 77.5%. The per-length results and generation protocol are in Appendix B.

Table 1: **Model effects of the Cosh exponent table.** Each row compares Geo and EVQ-COSH within one matched protocol. Entries are mean perplexities over the indicated training replications; lower is better.

| Model             | Seeds | Train length | Eval. length | Geo   | EVQ-COSH    |
|-------------------|-------|--------------|--------------|-------|-------------|
| 432M MLA          | 3     | 8K           | 8K           | 35.4  | 35.8        |
|                   |       |              | 16K          | 138.8 | <b>95.6</b> |
| 750M continuation | 1     | 4K           | 4K           | 22.0  | 22.3        |
|                   |       |              | 16K          | 45.1  | <b>24.4</b> |

**Matched Llama-3-8B adaptation.** We compare Native and EVQ-COSH under the same 300-step rank-64 Q/K/V/O LoRA adaptation, using physical 8K sequences (Hu et al., 2022; Grattafiori et al., 2024). The evaluation comprises 24 frozen temporal text packs from three domains. Figure 3 shows the full curve: Native-LoRA/EVQ-COSH-LoRA PPL is 6.82/10.07 at 8K, 108.96/24.07 at 16K, and 991.48/127.91 at 32K. The long-context NLL improvement holds in every pack in both extended lengths.

Source interventions help interpret this result. On ten true-16K passkey cases, median target-block hit@16 rises from 18.75% to 64.06%. Removing decode attention to the remote gold block raises

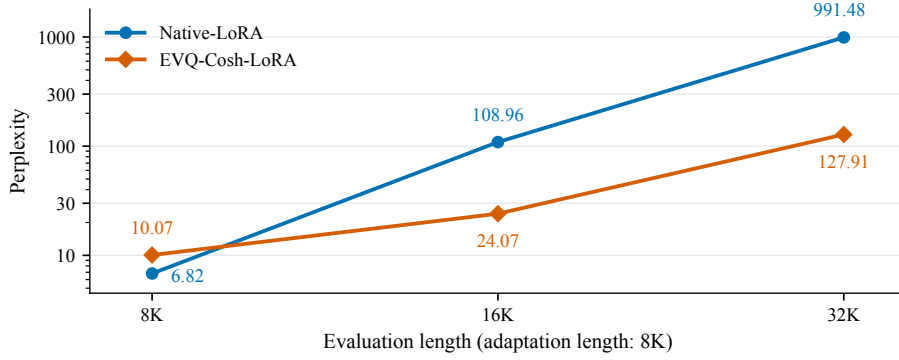


Figure 3: **The Cosh construction improves extrapolation under matched adaptation.** Llama-3-8B arms share an 8K, 300-step LoRA protocol and are evaluated on the same 24 text packs. The logarithmic PPL axis shows all three evaluation lengths. PPL is the exponential of the equal-domain mean NLL.

pooled answer-token NLL by 1.5055 in the EVQ-COSH arm, compared with  $-0.0095$  in Native. Thus the adapted model’s answer probability depends on the distant source. Appendix E.3 reports this intervention and the separate RULER-family continuation.

That continuation provides an autoregressive comparison: both arms receive the same 516 additional steps on 13 RULER families at physical length 8K, then generate answers on new rows. At 16K, the official RULER macro is 0.30% for Native-LoRA and 14.03% for EVQ-COSH-LoRA; at 8K it is 94.44/77.60%. The official task metrics allow partial credit; normalized whole-response exact match at 16K is 0/1.54%. These scores measure length transfer after matched task-family supervision.

**Composition with range extension.** In the 454M three-seed experiment, applying the same fixed-index smooth-ramp scaler to both trained tables gives 16K PPL of 157.7 for Geo and 107.5 for EVQ-COSH. The training-time allocation thus remains useful after this shared inference-time transformation. Appendix B.3 gives the operator, training protocol, and complete comparison.

**Further settings.** The same-initialization OLMo scale check and video-DiT comparison are reported with their full protocols in Apps. E.1 and B.6.

## 6 ADJUSTING EXPONENTS IN FROZEN MODELS

The frozen crossings in §3.3 show that trained weights use a particular exponent basis. This motivates adjustments defined relative to the checkpoint’s native table. We compare how these adjustments place the available exponent shifts within the existing rotary pairs.

### 6.1 A COMMON COORDINATE FOR EXPONENT ADJUSTMENTS

For native frequencies  $\omega_k^N$  and an installed table  $\omega'_k$ , define

$$d_k = \log \frac{\omega_k^N}{\omega'_k}, \quad \omega'_k = \omega_k^N e^{-d_k}, \quad \phi'_k - \phi_k^N = \frac{d_k}{\log b}. \quad (9)$$

Several familiar constructions then have explicit displacement profiles:

$$d_k = \begin{cases} -\log(1 - w_k + w_k/s), & \text{frequency blend,} \\ m_k \log s, & \text{log-frequency shift,} \\ \sum_{j < k} \log \lambda_j, & \text{mixed-radix conversion.} \end{cases} \quad (10)$$

Here  $s$  is the extension factor,  $w_k$  is a blend weight, and  $m_k$  is the normalized cumulative exponent shift. YaRN’s NTK-by-parts frequency map uses the first construction; MrRoPE uses the third (Peng et al., 2024; Tian et al., 2026). The coordinate lets us compare their realized exponent shapes directly.

Table 2: **Frozen-model RULER scores (%) at matched exponent range.** OLMo uses a held-out nine-task confirmation at 16K; Qwen uses a four-task development panel at 64K, with 20 rows per task. Within each model only the interior allocation changes.

| Exponent allocation | OLMo-1B      | Qwen-1.5B    |
|---------------------|--------------|--------------|
| Geometric           | 0.56         | 57.75        |
| Coarse ramp         | <b>61.04</b> | 64.00        |
| Residual-guided     | 60.47        | <b>66.50</b> |

Table 3: **A static extension across the native and extended windows.** Qwen-0.5B RULER-13 macro (%);  $g$  is the fixed cosine/sine amplitude used by each method.

| Installed table    | $g$    | 32K   | 64K          |
|--------------------|--------|-------|--------------|
| Native             | 1.0000 | 54.78 | 22.05        |
| YaRN-2             | 1.0693 | 55.94 | 45.37        |
| Normalized-index-2 | 1.0513 | 55.92 | <b>51.46</b> |

## 6.2 FIXED-RANGE AND SINGLE-TABLE COMPARISONS

**Changing only the normalized exponent allocation.** On frozen OLMo-2-1B and Qwen-2.5-1.5B, we hold the endpoints, 64 pairs, attention amplitude, inputs, and decoder fixed. A geometric allocation, a residual-guided allocation, and its nearest coarse ramp give the results in Table 2. The residual-guided construction moves pairs according to their remaining energy after projection onto the other sine–cosine pairs; the ramp approximates that profile without task labels. Both structured profiles produce large gains over uniform exponents. Their definitions and reference Native/YaRN rows are in Appendix E.2.

**One static table at both lengths.** A frozen normalized-index profile transfers a reference displacement curve onto the native slot grid and installs  $\omega'_k = \omega_k^N s^{-m_k}$ . Its placement can be compared with direct evaluation from local frequency gaps. On a new-seed Gemma-1.1-2B comparison with 128 rotary pairs (Gemma Team et al., 2024), normalized-index interpolation scores 79.00% versus 72.81% at 16K. Both arms share the frequency endpoints, amplitude, and 320 inputs; the paired improvement is 6.19 points with a 95% bootstrap interval [2.81, 9.63]. Appendix F.2 gives the two constructions and the corresponding 32-pair comparison.

On Qwen-2.5-0.5B, a new-seed 13-task RULER evaluation uses 20 examples per task at 32K and 64K. Every request uses the same  $s = 2$  table throughout prefill and decoding, including requests within the native window. The installed table and amplitude together give the comparison in Table 3: 6.09 points above YaRN at 64K, with a 95% paired task-stratified bootstrap interval [2.76, 9.58] points. At 32K the two methods are nearly equal. Appendix F.1 gives the frozen profile and the full task breakdown.

## 6.3 REDISTRIBUTING THE INTERMEDIATE-BAND SHIFT

MrRoPE-Pro progressively distributes the cumulative shift across an intermediate frequency band. We compare it with a boundary-matched (BM) profile, which keeps the same band endpoints and total scale while smoothing the per-slot radix increments toward the unmodified outer bands. Both are explicit constructions of  $m_k$  in (10); their exact discrete formulas are given in Appendix F.3.

In an OLMo-1B six-task confirmation, BM improves the 16K macro from 2.78% for MrRoPE-Pro to 51.32%; MrRoPE-Uni scores 32.12%. The comparison extends to natural inputs: across 631 eligible long examples from five QA tasks, the task-equal F1 rises from 21.62% to 25.44%. The 95% paired bootstrap interval for the improvement is [1.32, 6.29] points. Every task has a positive mean difference (Fig. 4). All methods use a fixed factor-four table; the input and decoding details are in Appendix F.5.

The three-model comparison reveals distinct preferences for the intermediate allocation (Table 4). BM gives the higher OLMo scores and Qwen-3B’s higher 32K score; MrRoPE-Pro gives the higher 128K

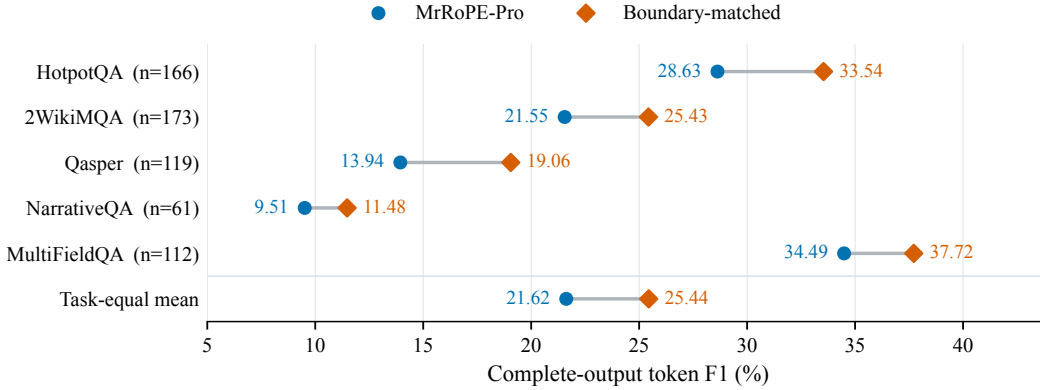


Figure 4: **An intermediate-band redistribution improves OLMo natural QA.** Complete-output token F1 on untruncated long inputs, using the same  $s = 4$  configuration and decoder in both arms. Sample counts are shown by task; the final row averages the five task means.

Table 4: **Matched six-task RULER comparisons (%)**. Each length column averages six task scores. Both methods use static  $s = 4$ ; the row counts refer to short/long prompts per arm.

| Model   | Lengths | Rows  | Short |       | Long  |       |
|---------|---------|-------|-------|-------|-------|-------|
|         |         |       | MrPro | BM    | MrPro | BM    |
| OLMo-1B | 4/16K   | 24/48 | 37.85 | 81.81 | 2.78  | 51.32 |
| Qwen-3B | 32/128K | 12/24 | 87.22 | 91.67 | 78.13 | 70.83 |
| Qwen-7B | 32/128K | 6/12  | 83.33 | 80.00 | 84.44 | 71.11 |

scores on both Qwen checkpoints. With band boundaries and cumulative scale fixed, these differences connect the preferred exponent shape to the checkpoint and operating length. Appendix F.4 reports every task.

## 7 RELATED WORK

**Range, scaling, and exponent shape.** RoPE’s rotation operator (Su et al., 2024) admits many frequency tables. Scaling-law and base-bound analyses study how the geometric family’s range affects length generalization (Liu et al., 2024; Xu et al., 2024). FMRoPE relates the learned frequency band to the base and training length, and chooses a base accordingly (Oka et al., 2026b). In that family the normalized exponents remain equally spaced; our fixed-range comparisons vary their interior distribution.

Position interpolation scales positions uniformly, while YaRN blends frequencies across bands (Chen et al., 2023a; Peng et al., 2024). LongRoPE combines dimension-wise scaling with a token-position threshold (Ding et al., 2024; Shang et al., 2025). MrRoPE gives uniform and progressive intermediate-band rules through mixed-radix conversion (Tian et al., 2026). Equation (10) expresses these frequency factors as exponent displacements. Our second stage compares their shapes; the first derives an exponent density and tests it at fixed frequency endpoints.

**Learning and using frequency bands.** LeRoPE learns one log-space scale per pair, shared across layers and heads; AdaRoPE learns head-specific frequencies and attention scaling (Karypis et al., 2026; Wang et al., 2026). Both optimize entries that can be expressed as exponent changes. Our contribution lies in their allocation geometry, explicit density construction, and controlled separation from range.

Frequency-use studies connect fast and slow bands to positional and symbolic attention, retrieval, and data-dependent scales (Barbero et al., 2025; Urrutia et al., 2026; Chiang & Yogatama, 2025; Oka et al., 2026a; Wu et al., 2026). DoPE uses truncated matrix entropy to identify low-rank structures in

rotated activations and selectively modifies positional encoding (Xiong et al., 2025). Our canonical correlations instead describe the supplied sine-cosine subspaces as their exponents vary.

**Changing the positional operator.** FoPE introduces Fourier components, while HoPE and clipped encodings alter the treatment of slow rotations (Hua et al., 2025; Chen et al., 2025; Li et al., 2026a). GRAPE generalizes positional transformations through group actions; Selective RoPE learns input-dependent rotations (Zhang et al., 2026; Movahedi et al., 2026). RePo learns contextual token positions before applying the encoding (Li et al., 2026b). These works change the positional map or its dependence on content. Our construction retains the standard rotation and designs its exponent table. Recent analyses of long-context positional and token confusion further motivate studying the interaction between frequency choice and the representations it acts on (Du et al., 2026).

## 8 DISCUSSION

The exponent distribution determines how a finite RoPE table samples positional scales. Fixed-range training establishes a behavioral effect of its interior shape, while range retargeting and weights-by-table crossings show that this effect interacts with the installed range and learned coefficients. The full sine-cosine analysis makes the underlying positional basis explicit and explains why slow exponents can supply strongly correlated directions.

These findings support two forms of design. Before or during learning, a specified allocation criterion can generate an explicit table, as illustrated by the Cosh construction and its trained-model results. After learning, adjustments relative to the native table provide a way to preserve its coordinate structure while reallocating the extension across pairs. The static-profile and boundary-matched experiments show practical gains, as well as checkpoint-dependent preferences for the adjustment shape.

A natural next step is to study exponent allocation under sparse attention. Selective token access and compressed representations change which positional distinctions reach the reader. The framework developed here supplies a way to examine which exponent distributions serve those architectures.

### ETHICS STATEMENT

This work introduces no new dataset, deployed system, or human-subject component. The proposed training-time construction changes only the RoPE inverse-frequency table and adds no learned parameters; the mature-checkpoint studies use the frozen-table or LoRA interventions stated in their protocols. The table construction remains inexpensive to audit, ablate, or revert. All models, corpora, and evaluation suites used here are public research artefacts used within their stated licences.

A better allocated frequency table may reduce the adaptation or inference effort required to reach a target context range, and the same mechanism could be used to extend the context capability of language or video models, with the general risks associated with stronger generative models. Our evaluations measure positional and long-context endpoints, not factual reliability or safety; deployment claims require application-specific evaluation.

### REPRODUCIBILITY STATEMENT

The appendices specify the installed frequency tables, exponent grids, attention amplitudes, training and adaptation recipes, datasets, and evaluation procedures. The fixed-range experiment reports three paired training seeds; the larger-model and frozen-model tables identify their comparison units and sample counts. The accompanying source archive contains the active TeX, bibliography, plotted figures, numerical summaries, a SHA256 manifest, and a standalone verification of the explicit Gram-matrix examples. The appendices provide the derivations and exponent-adjustment formulas. Model checkpoints and training-data streams are separate from this manuscript archive; its build reproduces the paper, and its numerical records support inspection of the reported comparisons.

## AI USE STATEMENT

We used generative AI tools to help develop conceptual frameworks and mathematical claims; propose and refine hypotheses; draft and check proofs; give feedback on methodology and experiments; implement methods and deterministic data-reformatting and analysis code; translate or rephrase research notes; and interpret results. We did not use generative AI to generate dataset content or to perform qualitative or thematic data analysis; survey, interview, and transcription tasks do not apply to this work.

For tasks for which disclosure is recommended, we also used these tools to suggest experimental parameters; write, edit, and test software and artefacts; create or modify figures; draft, edit, and structure parts of the manuscript; brainstorm and identify research gaps; search for and summarise literature; and format references. Every empirical quantity reported here comes from executed code over recorded run artefacts rather than language-model generation.

All AI-assisted work was reviewed before inclusion. We take responsibility for the final text, claims, proofs, code, figures, and experimental artefacts.

## REFERENCES

- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 3119–3137, 2024. doi: 10.18653/v1/2024.acl-long.172. URL <https://aclanthology.org/2024.acl-long.172/>.
- Federico Barbero, Alex Vitvitskyi, Christos Perivolaropoulos, Razvan Pascanu, and Petar Veličković. Round and round we go! what makes rotary positional encodings useful? In *International Conference on Learning Representations*, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/hash/e6d58fc68c0f3c36ae6e0e64478a69c0-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2025/hash/e6d58fc68c0f3c36ae6e0e64478a69c0-Abstract-Conference.html).
- Sidney Black, Stella Biderman, Eric Hallahan, Quentin Anthony, Leo Gao, Laurence Golding, Horace He, Connor Leahy, Kyle McDonell, Jason Phang, Michael Pieler, Usven Sai Prashanth, Shivanshu Purohit, Laria Reynolds, Jonathan Tow, Ben Wang, and Samuel Weinbach. GPT-NeoX-20B: An open-source autoregressive language model. In *Proceedings of BigScience Episode #5 – Workshop on Challenges & Perspectives in Creating Large Language Models*, pp. 95–136, 2022. doi: 10.18653/v1/2022.bigscience-1.9. URL <https://aclanthology.org/2022.bigscience-1.9/>.
- Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation. *arXiv preprint arXiv:2306.15595*, 2023a. doi: 10.48550/arXiv.2306.15595. URL <https://arxiv.org/abs/2306.15595>.
- Yuhan Chen, Ang Lv, Jian Luan, Bin Wang, and Wei Liu. HoPE: A novel positional encoding without long-term decay for enhanced context awareness and extrapolation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 23044–23056, 2025. doi: 10.18653/v1/2025.acl-long.1123. URL <https://aclanthology.org/2025.acl-long.1123/>.
- Yukang Chen, Shengju Qian, Haotian Tang, Xin Lai, Zhijian Liu, Song Han, and Jiaya Jia. Longlora: Efficient fine-tuning of long-context large language models. *arXiv preprint arXiv:2309.12307*, 2023b. doi: 10.48550/arXiv.2309.12307. URL <https://arxiv.org/abs/2309.12307>.
- Ting-Rui Chiang and Dani Yogatama. The rotary position embedding may cause dimension inefficiency in attention heads for long-distance retrieval. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 13552–13562, 2025. doi: 10.18653/v1/2025.findings-acl.697. URL <https://aclanthology.org/2025.findings-acl.697/>.
- DeepSeek-AI. DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434*, 2024a. doi: 10.48550/arXiv.2405.04434. URL <https://arxiv.org/abs/2405.04434>.

- DeepSeek-AI. DeepSeek-V3 technical report. *arXiv preprint arXiv:2412.19437*, 2024b.
- Yiran Ding, Li Lyna Zhang, Chengruidong Zhang, Yuanyuan Xu, Ning Shang, Jiahang Xu, Fan Yang, and Mao Yang. LongRoPE: Extending LLM context window beyond 2 million tokens. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, pp. 11091–11104, 2024. URL <https://proceedings.mlr.press/v235/ding24i.html>.
- Yufeng Du, Phillip Harris, Minyang Tian, Eliu A. Huerta, Srikanth Ronanki, Subendhu Rongali, Aram Galstyan, and Hao Peng. RoPE distinguishes neither positions nor tokens in long contexts, provably. *arXiv preprint arXiv:2605.15514*, 2026. doi: 10.48550/arXiv.2605.15514. URL <https://arxiv.org/abs/2605.15514>.
- Gemma Team et al. Gemma: Open models based on gemini research and technology. *arXiv preprint arXiv:2403.08295*, 2024. URL <https://arxiv.org/abs/2403.08295>.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. doi: 10.48550/arXiv.2407.21783. URL <https://arxiv.org/abs/2407.21783>.
- Robert M. Gray and David L. Neuhoff. Quantization. *IEEE Transactions on Information Theory*, 44(6):2325–2383, 1998. doi: 10.1109/18.720541.
- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pp. 6609–6625, 2020. doi: 10.18653/v1/2020.coling-main.580. URL <https://aclanthology.org/2020.coling-main.580/>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>.
- Ermo Hua, Che Jiang, Xingtai Lv, Kaiyan Zhang, Youbang Sun, Yuchen Fan, Xuekai Zhu, Biqing Qi, Ning Ding, and Bowen Zhou. Fourier position embedding: Enhancing attention’s periodic extension for length generalization. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 24932–24949, 2025. URL <https://proceedings.mlr.press/v267/hua25b.html>.
- Petros Karypis, Sean O’Brien, Shreyas Kadekodi, Rui Zhu, and Julian McAuley. LeRoPE: Learnable RoPE frequencies improve language modeling. *arXiv preprint arXiv:2607.10134*, 2026. doi: 10.48550/arXiv.2607.10134. URL <https://arxiv.org/abs/2607.10134>.
- Haoran Li, Sucheng Ren, Alan Yuille, and Feng Wang. CoPE: Clipped RoPE as a scalable free lunch for long context LLMs. *arXiv preprint arXiv:2602.05258*, 2026a. doi: 10.48550/arXiv.2602.05258. URL <https://arxiv.org/abs/2602.05258>.
- Huayang Li, Tianyu Zhao, Deng Cai, and Richard Sproat. RePo: Language models with context re-positioning. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026b. URL <https://icml.cc/virtual/2026/poster/64002>. arXiv:2512.14391.
- Xiaoran Liu, Hang Yan, Chenxin An, Xipeng Qiu, and Dahua Lin. Scaling laws of RoPE-based extrapolation. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://openreview.net/forum?id=J07k0SJ5V6>. arXiv:2310.05209.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations (ICLR)*, 2017. URL <https://openreview.net/forum?id=Byj72udxe>. arXiv:1609.07843.
- Sajad Movahedi, Timur Carstensen, Arshia Afzal, Frank Hutter, Antonio Orvieto, and Volkan Cevher. Selective rotary position embedding. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=AQo1SEElNb>.

- Yui Oka, Kentaro Hanafusa, Taku Hasegawa, Kyosuke Nishida, and Kuniko Saito. Probing rotary position embeddings through frequency entropy. In *International Conference on Learning Representations (ICLR)*, 2026a. URL <https://openreview.net/forum?id=1JZuEDq62N>.
- Yui Oka, Itsumi Saito, Kyosuke Nishida, and Kuniko Saito. Frequency bands in RoPE: Base frequency and context length shape the interpolation–extrapolation trade-off. In *International Conference on Learning Representations (ICLR)*, 2026b. URL <https://openreview.net/forum?id=PR1PPxvG9Q>.
- Guilherme Penedo, Hynek Kydlíček, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro von Werra, and Thomas Wolf. The FineWeb datasets: Decanting the web for the finest text data at scale. *arXiv preprint arXiv:2406.17557*, 2024. doi: 10.48550/arXiv.2406.17557. URL <https://arxiv.org/abs/2406.17557>.
- Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. YaRN: Efficient context window extension of large language models. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=wHBfxhZulu>.
- Ning Shang, Li Lyna Zhang, Siyuan Wang, Gaokai Zhang, Gilsinia Lopez, Fan Yang, Weizhu Chen, and Mao Yang. LongRoPE2: Near-lossless LLM context window scaling. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 54203–54218, 2025. URL <https://proceedings.mlr.press/v267/shang25a.html>.
- Nitish Srivastava, Elman Mansimov, and Ruslan Salakhutdinov. Unsupervised learning of video representations using LSTMs. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *PMLR*, pp. 843–852, 2015. URL <https://proceedings.mlr.press/v37/srivastava15.html>.
- Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.
- Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, Allyson Ettinger, Michal Guerquin, David Heineman, Hamish Ivison, Pang Wei Koh, Jiacheng Liu, Saumya Malik, William Merrill, Lester James V. Miranda, Jacob Morrison, Tyler Murray, Crystal Nam, Jake Poznanski, Valentina Pyatkin, Aman Rangapur, Michael Schmitz, Sam Skjonsberg, David Wadden, Christopher Wilhelm, Michael Wilson, Luke Zettlemoyer, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*, 2025. doi: 10.48550/arXiv.2501.00656. URL <https://arxiv.org/abs/2501.00656>.
- Qingyuan Tian, Wenhong Zhu, Xiaoran Liu, Xiaofeng Wang, and Rui Wang. MrRoPE: Mixed-radix rotary position embedding. In *International Conference on Learning Representations (ICLR)*, 2026. URL [https://proceedings.iclr.cc/paper\\_files/paper/2026/hash/69413f87e5a34897cd010ca698097d0a-Abstract-Conference.html](https://proceedings.iclr.cc/paper_files/paper/2026/hash/69413f87e5a34897cd010ca698097d0a-Abstract-Conference.html).
- Felipe Urrutia, Jorge Salas, Alexander Kozachinskiy, Cristian Buc Calderon, Hector Pasten, and Cristobal Rojas. Decoupling positional and symbolic attention behavior in transformers. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=V38yAoqddQ>.
- Shaowen Wang, Yuke Zheng, Tansheng Zhu, Shuang Chen, Shaofan Liu, Suncong Zheng, and Jian Li. AdaRoPE: Not all attention heads should rotate and scale equally. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/60704>. arXiv:2607.19363.
- Xilin Wei, Xiaoran Liu, Yuhang Zang, Xiaoyi Dong, Pan Zhang, Yuhang Cao, Jian Tong, Haodong Duan, Qipeng Guo, Jiaqi Wang, Xipeng Qiu, and Dahua Lin. VideoRoPE: What makes for good video rotary position embedding? In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 66118–66136, 2025. URL <https://proceedings.mlr.press/v267/wei25h.html>.

- Xinyi Wu, Siyuan Liu, and Ali Jadbabaie. How data shapes RoPE frequency usage: From positional scale matching to length generalization. *arXiv preprint arXiv:2607.07678*, 2026. doi: 10.48550/arXiv.2607.07678. URL <https://arxiv.org/abs/2607.07678>.
- Jing Xiong, Liyang Fan, Hui Shen, Zunhai Su, Min Yang, Lingpeng Kong, and Ngai Wong. DoPE: Denoising rotary position embedding. *arXiv preprint arXiv:2511.09146*, 2025. doi: 10.48550/arXiv.2511.09146. URL <https://arxiv.org/abs/2511.09146>.
- Mingyu Xu, Xin Men, Bingning Wang, Qingyu Zhang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. Base of RoPE bounds context length. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2773. URL [https://proceedings.neurips.cc/paper\\_files/paper/2024/hash/9f12dd32d552f3ad9eaa0e9dfec291be-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2024/hash/9f12dd32d552f3ad9eaa0e9dfec291be-Abstract-Conference.html).
- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. doi: 10.48550/arXiv.2412.15115. URL <https://arxiv.org/abs/2412.15115>.
- Yifan Zhang, Zixiang Chen, Yifeng Liu, Zhen Qin, Huizhuo Yuan, Kangping Xu, Yang Yuan, Quanquan Gu, and Andrew Chi-Chih Yao. Group representational position encoding. In *International Conference on Learning Representations (ICLR)*, 2026. URL [https://proceedings.iclr.cc/paper\\_files/paper/2026/file/5cb58625f49ddf70fe2d527e9e4bbae5-Paper-Conference.pdf](https://proceedings.iclr.cc/paper_files/paper/2026/file/5cb58625f49ddf70fe2d527e9e4bbae5-Paper-Conference.pdf).
- Min Zhao, Guande He, Yixiao Chen, Hongzhou Zhu, Chongxuan Li, and Jun Zhu. RIFLEx: A free lunch for length extrapolation in video diffusion transformers. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 77539–77557, 2025. URL <https://proceedings.mlr.press/v267/zhao25m.html>.

## A PROOF DETAILS

### A.1 SUBSPACE GEOMETRY: CROSS-GRAM, BUDGET IDENTITY, AND COLLAPSE

**Closed-form cross-Gram.** For  $\Delta \sim \text{Unif}[0, L]$  and  $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$ , write  $d = (\omega - \nu)L$ ,  $s = (\omega + \nu)L$ ,  $a(t) = \sin t/t$  and  $b_\star(t) = (1 - \cos t)/t$ . Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_\star(s) - b_\star(d) \\ b_\star(s) + b_\star(d) & a(d) - a(s) \end{bmatrix}, \quad (11)$$

and  $S_\omega = H_{\omega\omega}$ . Since  $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$  is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and  $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$  is invariant under  $x_\omega \mapsto x_\omega O$  for any invertible  $O$  (in particular under rotation of the phase inside a pair), because  $S_\omega$  transforms congruently and the whitening cancels  $O$ .

**Basis invariance.** The cancellation can be stated exactly. Under independent invertible basis changes  $x_\omega \mapsto x_\omega A_\omega$  and  $x_\nu \mapsto x_\nu A_\nu$ , let  $U_j = S_j^{1/2} A_j (A_j^\top S_j A_j)^{-1/2}$  for  $j \in \{\omega, \nu\}$ . Then  $U_j^\top U_j = I$ , and the transformed whitened cross-Gram is  $\tilde{Q}_{\omega\nu} = U_\omega^\top Q_{\omega\nu} U_\nu$ . Left and right orthogonal factors preserve the two canonical correlations and therefore  $c_{\omega\nu}$ . Thus  $c_{\omega\nu}$  is an intrinsic measure of overlap between the two positional subspaces.

**Theorem 3** (Spectral budget identity). *For the block-whitened Gram matrix  $\Gamma$  with diagonal blocks  $I_2$ ,  $\text{tr } \Gamma = 2K$  and  $\text{tr}(\Gamma^2) = 2K[1 + (K-1)\bar{c}]$ . Hence its Rényi-2 effective rank is given by (5).*

*Proof of Theorem 3.*  $\Gamma$  is the block matrix with blocks  $\Gamma_{ij} = Q_{\omega_i \omega_j}$  and  $\Gamma_{ii} = I_2$ . Hence  $\text{tr } \Gamma = \sum_{i=1}^K \text{tr } I_2 = 2K$ . For the second moment,  $\text{tr}(\Gamma^2) = \sum_{i,j} \text{tr}(\Gamma_{ij} \Gamma_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K-1)\bar{c}$ , using  $\Gamma_{ji} = \Gamma_{ij}^\top$  and the definition  $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$  with  $\|Q_{ij}\|_F^2 = 2c_{ij}$ . Dividing gives (5).  $\square$

The identity determines Rényi-2 rank exactly from the mean pairwise redundancy. Shannon effective rank and log-determinant retain higher-order multi-subspace dependence on the full spectrum.

*Proof of Proposition 1.* Put  $t = \Delta/L$ ,  $x = \omega L$ , and use  $\theta = x^2$  as the local parameter. The rescaled pair has the analytic basis

$$u_1(\theta, t) = \cos(\sqrt{\theta} t) = 1 - \frac{1}{2}\theta t^2 + O(\theta^2), \quad (12)$$

$$u_2(\theta, t) = \frac{\sin(\sqrt{\theta} t)}{\sqrt{\theta}} = t - \frac{1}{6}\theta t^3 + O(\theta^2). \quad (13)$$

Thus  $V_\omega \rightarrow V_0 = \text{span}\{1, \Delta\}$ . The fourth-order coefficient follows by retaining the part of the first perturbation transverse to  $V_0$ . Under the uniform measure on  $t \in [0, 1]$ ,

$$\Pi_0^\perp t^2 = p_2(t) := t^2 - t + \frac{1}{6}, \quad (14)$$

$$\Pi_0^\perp t^3 = p_3(t) := t^3 - \frac{9}{10}t + \frac{1}{5}, \quad (15)$$

where  $\Pi_0^\perp$  is the  $L_2[0, 1]$  projection off  $\text{span}\{1, t\}$ . In the base  $[1, t]$ , its Gram and the residual perturbation Gram are

$$G_0 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}, \quad M = \begin{bmatrix} \langle -p_2/2, -p_2/2 \rangle & \langle -p_2/2, -p_3/6 \rangle \\ \langle -p_3/6, -p_2/2 \rangle & \langle -p_3/6, -p_3/6 \rangle \end{bmatrix} = \begin{bmatrix} 1/720 & 1/1440 \\ 1/1440 & 1/2800 \end{bmatrix}. \quad (16)$$

The squared chordal distance between the two nearby planes is therefore

$$2 - \|Q_{x,y}\|_F^2 = (x^2 - y^2)^2 \text{tr}(G_0^{-1} M) + O(\epsilon^6) = \frac{19}{12600} (x^2 - y^2)^2 + O(\epsilon^6), \quad (17)$$

with  $\epsilon = \max\{|x|, |y|\} \rightarrow 0$ . The exact coefficient is reproduced with rational polynomial inner products by `scripts/analysis/full_rope_collision_audit.py`; at  $x=0.05, y=0.10$  its analytic Gram gives exact/leading ratio 1.00058.

For the softmax metric with fixed  $p$ ,  $F = \text{diag}(p) - pp^\top$  satisfies  $F\mathbf{1} = 0$ , so the constant direction is annihilated before whitening. Writing  $\bar{h} = h - \mathbb{E}_p h$ , the same expansion gives

$$\frac{\overline{\sin(\omega\Delta)}}{\omega} \longrightarrow \Delta - \mathbb{E}_p \Delta, \quad -\frac{\overline{2\cos(\omega\Delta)}}{\omega^2} \longrightarrow \Delta^2 - \mathbb{E}_p \Delta^2. \quad (18)$$

These functions span the centred limit whenever  $p$  has nondegenerate support on at least three distances. The separation weights  $p$  therefore determine which polynomial directions survive centering.  $\square$

**Directional overlap and feature scale.** For the original, unwhitened pair on  $[0, L]$ , put  $x = \omega L$  and  $\text{sinc } x = \sin x/x$ . The self-Gram eigenvalues are

$$\lambda_{\pm}(S_{\omega}) = \frac{1 \pm |\text{sinc } x|}{2}, \quad \lambda_{-}(S_{\omega}) \sim \frac{x^2}{12}, \quad \kappa(S_{\omega}) \sim \frac{12}{x^2} \quad (x \rightarrow 0).$$

Thus the slow pair has a poorly scaled second direction before normalization. Block whitening removes this within-pair scaling and measures overlap between positional directions. Raw feature energy, coefficient magnitude, and numerical precision determine how strongly those directions can contribute to a model.

The standard  $u_k = k/K$  grid gives 23 slow pairs in Figure 5; the endpoint-inclusive grid used in Figure 6 gives 24 at  $K = 64$ . Both figures use the uniform separation measure.

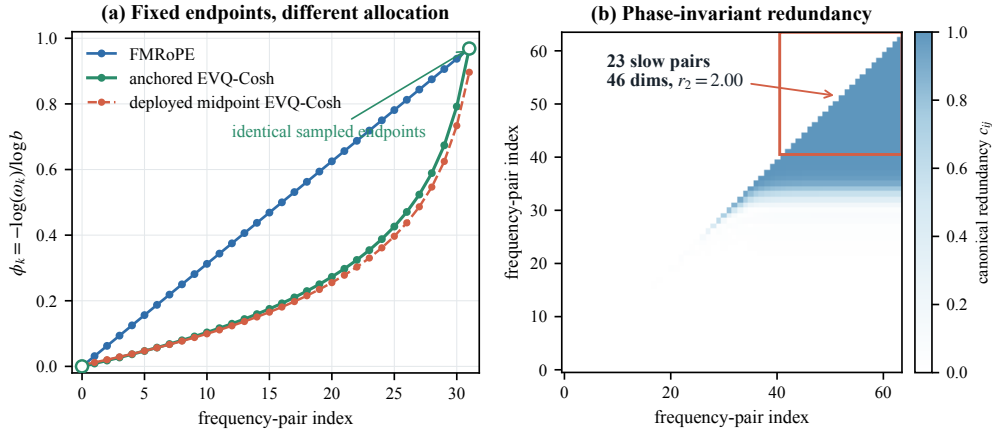


Figure 5: **Where a finite frequency table spends its budget.** (a) At  $K=32$ , Cosh concentrates interior channels toward the fast end while endpoint normalisation preserves sampled support; the deployed midpoint table is shown separately. (b) For standard RoPE at  $L=4096$ ,  $b=5 \times 10^5$ , and  $K=64$ , phase-invariant full-subspace redundancy concentrates in the 23 pairs with  $\omega L \leq 1$ .

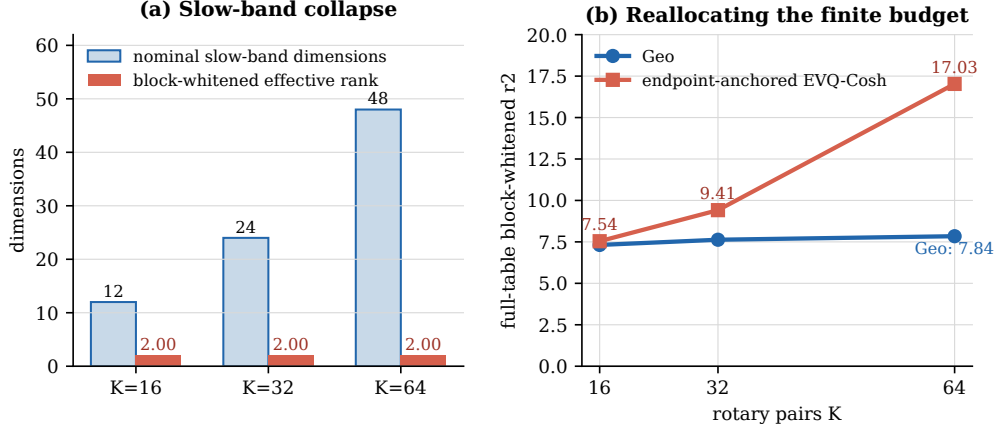


Figure 6: **Spectral-budget anatomy across channel counts.** Panel (a) uses the endpoint-inclusive diagnostic grid at  $L=4096$  and  $b=500K$ : as  $K$  grows from 16 to 64, the nominal dimension assigned to pairs with  $\omega L \leq 1$  grows from 12 to 48, while their phase-invariant effective rank remains approximately two. Panel (b) evaluates the complete fixed-endpoint table under the same uniform separation prior, using the zero-search reference  $\tau=2K/\sqrt{L}$ ; anchored EVQ-COSH spends progressively more of the available full-pair dimension than Geo as the budget grows.

## A.2 EFFECTIVE DIMENSION AND MULTISCALE COVERAGE

The budget identity is an exact account of positional dimension under its separation prior. A second requirement is where that dimension is placed across scales. The distinction is already visible in an exact construction that also identifies the complete parity class omitted by the usual  $2\pi/L$  Fourier description.

**Proposition 4** (Parity-lattice orthogonality and recurrence). *Let  $a_1, \dots, a_K$  be distinct positive integers of one parity and set*

$$\omega_k = \frac{\pi a_k}{L}. \quad (19)$$

*Under  $\Delta \sim \text{Unif}[0, L]$ , the  $K$  full sin/cos subspaces are exactly mutually orthogonal: their block-whitened Gram is  $\Gamma = I_{2K}$  and  $r_2(\Gamma) = 2K$ . If support is free but restricted to  $0 < \omega \leq 1$ , the largest same-parity class in this construction has*

$$K_{\text{par}}(L) = \left\lceil \frac{\lfloor L/\pi \rfloor}{2} \right\rceil \quad (20)$$

*pairs. The resulting feature map is recurrent: for even  $a_k$ ,  $\Phi_\Omega(\Delta + L) = \Phi_\Omega(\Delta)$ ; for odd  $a_k$ ,  $\Phi_\Omega(\Delta + L) = -\Phi_\Omega(\Delta)$  and hence  $\Phi_\Omega(\Delta + 2L) = \Phi_\Omega(\Delta)$ .*

*Proof.* For  $i \neq j$ , same parity makes both  $a_i - a_j$  and  $a_i + a_j$  nonzero even integers. Consequently

$$(\omega_i \mp \omega_j)L = \pi(a_i \mp a_j) \in 2\pi\mathbb{Z}, \quad (21)$$

so every entry of the cross-Gram in (11) vanishes:  $a((\omega_i \mp \omega_j)L) = 0$  and  $b_*((\omega_i \mp \omega_j)L) = 0$ . Likewise  $2\omega_i L = 2\pi a_i$  gives  $S_{\omega_i} = \frac{1}{2}I_2$ . Thus the unwhitened Gram is  $\frac{1}{2}I_{2K}$  and block whitening gives  $\Gamma = I_{2K}$ . Among the integers  $1, \dots, \lfloor L/\pi \rfloor$ , the larger parity class has  $\lceil \lfloor L/\pi \rfloor / 2 \rceil$  elements, proving (20) for this construction. Finally,  $e^{i\omega_k(\Delta+L)} = (-1)^{a_k} e^{i\omega_k \Delta}$ ; the common parity gives the stated period or antiperiod, and every antiperiodic table repeats after  $2L$ .  $\square$

Proposition 4 couples maximal rank on  $[0, L]$  with recurrence beyond that interval. Its orthogonality also depends on the separation measure: weighting the same lattice by the triangular causal histogram introduces off-diagonal Gram terms. Thus interval coverage and pairwise rank capture different features of a positional basis.

### A.3 TWO STATIC ORDERING COUNTEREXAMPLES

Two explicit examples show how phase coupling and interval length change the ordering of frequency tables.

**Cosine-only collision can choose the lower-rank table.** Take  $K = 4$ ,  $L = 8\pi$ , and the two descending frequency tables

$$\Omega_A = (1, 5/8, 3/8, 1/4), \quad \Omega_B = (1, 6.01/8, 4.01/8, 1/4).$$

They share both endpoints. Define  $C_{\cos}$  as the mean squared normalized cosine-cosine overlap over distinct pairs. For  $\Omega_A$ , every cosine is an integer harmonic on  $[0, L]$ , so  $C_{\cos}(A) = 0$  exactly. Mixed harmonic parities nevertheless produce nonzero sine-cosine cross terms. For  $\Omega_B$ , the small perturbation of an even-harmonic table gives

$$C_{\cos}(A) = 0 < 1.38725 \times 10^{-5} = C_{\cos}(B), \quad (22)$$

$$r_2(A) = 6.29478 < 7.99971 = r_2(B). \quad (23)$$

Thus a strict cosine-only preference can have the opposite full-subspace rank ordering. Both values follow directly from Eq. (11).

**Full-subspace collision can reverse across lengths.** Now use  $K = 4$ ,  $L = 2\pi$ , and the fixed tables

$$\Omega_A = (1, 0.99, 0.02, 0.01), \quad \Omega_B = (1, 2/3, 1/3, 0.01).$$

Again the endpoints agree. Writing  $C_M$  for mean phase-invariant collision under the uniform measure on  $[0, M]$ , Eq. (11) gives

$$C_L(A) = 0.538996 < 0.631721 = C_L(B), \quad (24)$$

$$C_{2L}(A) = 0.384445 > 0.204995 = C_{2L}(B), \quad (25)$$

$$C_{4L}(A) = 0.345188 > 0.033506 = C_{4L}(B). \quad (26)$$

The close pairs in  $A$  retain high overlap as the interval grows, while the more evenly spread frequencies in  $B$  become easier to distinguish. The example makes the interval dependence of the geometry explicit.

All frequency entries and measures are specified above. The accompanying `figs/verify_explicit_geometry.py` evaluates the closed-form Gram and checks it against independent 128-point Gauss-Legendre quadrature; the largest discrepancy in the displayed metrics is below  $4 \times 10^{-14}$ . These finite basis examples complement the trained allocation comparisons and the weight-table crossings.

### A.4 POST-HOC TRANSPLANT OBSTRUCTION

**Table 5: Weights co-adapt to their training-time frequency table.** Two seed-42 50M models trained with the geometric and EVQ-COSH tables, each evaluated under both runtime tables with all parameters frozen (1,920 head-query observations). Both runtime tables use the shared midpoint grid  $u_k = (k + 1/2)/K$  for this diagnostic. The matched pairs remain near PPL 7, whereas both cross-swaps fail; the worst cell is also the one whose static  $r_2$  improves most.

| Trained weights | Runtime table | PPL         | Rényi-2 rank $r_2$ |
|-----------------|---------------|-------------|--------------------|
| Geo             | Geo           | <b>7.14</b> | 4.57               |
| Geo             | EVQ-COSH      | 76.20       | <b>12.54</b>       |
| EVQ-COSH        | Geo           | 23.05       | 4.57               |
| EVQ-COSH        | EVQ-COSH      | <b>7.16</b> | 12.54              |

**Theorem 5** (Post-hoc transplant obstruction). *Let  $\mathcal{R}_\Omega(\Delta)$  be the block-rotation operator of a frequency multiset  $\Omega$ . If position-independent invertible maps  $A, B$  satisfy  $A^\top \mathcal{R}_{\Omega'}(\Delta)B = \mathcal{R}_\Omega(\Delta)$  on an open interval containing zero, then  $\Omega'$  and  $\Omega$  have the same frequency multiset up to sign and permutation. Repeated frequencies may mix within their full equal-frequency invariant subspace.*

*Proof of Theorem 5.* Both  $\mathcal{R}_\Omega$  and  $\mathcal{R}_{\Omega'}$  are block-diagonal with  $2 \times 2$  rotation blocks, hence  $\mathcal{R}_\Omega(0) = \mathcal{R}_{\Omega'}(0) = I$ . Evaluating the hypothesis at  $\Delta=0$  gives  $A^\top B = I$ , so  $B = A^{-\top}$  and the hypothesis becomes the similarity  $A^\top \mathcal{R}_{\Omega'}(\Delta) A^{-\top} = \mathcal{R}_\Omega(\Delta)$  on an interval. Each side is the exponential of a constant generator,  $\mathcal{R}_\Omega(\Delta) = \exp(\Delta G_\Omega)$  with  $G_\Omega = \bigoplus_k \omega_k J$ ,  $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ . Differentiating at  $\Delta=0$  gives  $A^\top G_{\Omega'} A^{-\top} = G_\Omega$ , so the generators are similar and share a spectrum. The spectrum of  $G_\Omega$  is  $\{\pm i\omega_k\}_{k \leq K}$  with multiplicity, so the frequency multisets agree up to sign and permutation. When a frequency is repeated, similarity may mix the entire equal-frequency invariant subspace, including across its original two-dimensional blocks. For integer positions the same argument is replaced by a one-step spectral argument: evaluating at  $\Delta = 0$  again gives  $B = A^{-\top}$ , while  $\Delta = 1$  makes  $\mathcal{R}_{\Omega'}(1)$  and  $\mathcal{R}_\Omega(1)$  similar. Their eigenvalue multisets  $\{e^{\pm i\omega_k}\}$  therefore agree, which identifies frequencies up to sign, permutation, and the  $2\pi$  alias.  $\square$

## A.5 PROOF OF THE CLOSED-FORM VARIATIONAL OPTIMUM

This subsection proves Theorem 2 directly from the stated variational objective. Starting from the kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (27)$$

the allocation functional is (6), which under the mass normalization  $\int_0^1 \rho = 1$  with Lagrange multiplier  $\nu$  has constrained first variation

$$\alpha \rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (28)$$

**Existence.** For rigor, view  $\mathcal{C}_{\text{app}}$  first as a functional on  $L^2([0, 1])$  and minimize over  $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e., } \int_0^1 \rho = 1\}$ . The  $\alpha \|\rho\|_2^2$  term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical  $C^2$  derivation is justified.

Direct differentiation of  $g$  gives

$$g'(\phi) = \int_\phi^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (29)$$

Differentiating the first variation twice in  $\phi$  eliminates  $\nu$  (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (30)$$

Evaluating the pre-differentiated stationarity together with  $g(0)=0$ ,  $g'(1)=0$ , and  $\int_0^1 \rho = 1$  gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (31)$$

since differentiating  $\alpha \rho(\phi) + \beta g(\phi) + \nu = 0$  once and evaluating at  $\phi=1$  gives  $\alpha \rho'(1) + \beta g'(1) = 0$  hence  $\rho'(1)=0$ , and at  $\phi=0$  gives  $\alpha \rho'(0) + \beta g'(0) = 0$  with  $g'(0) = \int_0^1 \rho = 1$ , so  $\rho'(0) = -\beta/\alpha = -\tau^2$ . Substituting  $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$  into (30) and (31) together with  $\int_0^1 \rho = 1$  gives the unique positive interior stationary density

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad (32)$$

which is main-text Theorem 2. Mass normalization enters the boundary condition through  $g'(0) = \int_0^1 \rho = 1$ . Positivity holds on the closed interval since  $\rho_\tau(\phi) \geq \rho_\tau(1) = \tau/\sinh \tau > 0$  for  $\tau > 0$  (with  $\rho_\tau(0) = \tau \coth \tau > 0$ ); the inequality constraint  $\rho > 0$  is therefore inactive throughout  $[0, 1]$  and the unconstrained Euler-Lagrange solution coincides with the KKT-constrained solution. For the degenerate case  $\beta=0$  the ODE reduces to  $\rho''=0$  with  $\rho'(0)=\rho'(1)=0$  and  $\int \rho=1$ , forcing  $\rho \equiv 1$ , which is also the  $\tau \rightarrow 0$  limit of (32).

**Convexity / PSD of  $\mathcal{C}_{\text{app}}$ .** For  $\alpha > 0$  and  $\beta \geq 0$ , the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal  $\alpha \int \rho^2$  is manifestly PSD on  $L^2$ , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left( \int_s^1 f(u) du \right)^2 ds \geq 0. \quad (33)$$

The strictly convex functional on  $L^2([0, 1])$  therefore has the displayed stationary density as its unique constrained minimizer.

The Cosh solution has an analytically invertible CDF, so its quantiles turn the continuous allocation directly into a finite frequency table.

#### A.6 SINGLE-CROSSING BUDGET SHIFT

**Lemma 6** (Single-crossing budget shift). *For every  $\tau > 0$ ,  $\rho_\tau$  crosses the uniform density exactly once, at*

$$\phi_c(\tau) = 1 - \tau^{-1} \operatorname{arcosh}(\sinh \tau / \tau) \leq 1 - 1/\sqrt{3}. \quad (34)$$

*Proof of Lemma 6.* For  $\tau > 0$ ,  $\rho_\tau(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$  is strictly decreasing on  $[0, 1]$ , with  $\rho_\tau(0) = \tau \coth \tau > 1$  and  $\rho_\tau(1) = \tau / \sinh \tau < 1$ . Hence the crossing is unique, and solving  $\rho_\tau(\phi_c) = 1$  gives (34). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n (2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is  $3^n \geq 2n + 1$ . Monotonicity of  $\operatorname{arcosh}$  yields  $\operatorname{arcosh}(\sinh \tau / \tau) \geq \tau / \sqrt{3}$ , proving the bound.  $\square$

#### A.7 SURROGATE SELF-CONSISTENCY THEOREM

The surrogate  $\mathcal{C}_{\text{app}}$  contains two integral terms  $T_1(\tau) = \int_0^1 \rho^2 d\phi$  and  $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi$ , where  $\rho(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$  is the normalized cosh density.

**Theorem 7** (Surrogate self-consistency). *For all  $\tau > 0$ :*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (35)$$

*Proof.* The Green's function  $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$  satisfies  $g''(\phi) = -\rho(\phi)$  with  $g(0) = 0$ ,  $g'(1) = 0$  (from the min kernel structure; see §A). Define  $h(\phi) = \rho(0) - \rho(\phi)$ . Then  $h'' = -\rho''$ , and under the pure-tether ODE  $\rho'' = \tau^2 \rho$  this gives  $h'' = -\tau^2 \rho$ , with  $h(0) = 0$  and  $h'(0) = -\rho'(0)$ .

Define  $f(\phi) = \tau^2 g(\phi) - h(\phi)$ . Then  $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$ , so  $f$  is linear in  $\phi$ . The boundary values give  $f(0) = \tau^2 g(0) - h(0) = 0$  and  $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$  (using  $g'(0) = \int_0^1 \rho = 1$  and  $\rho'(0) = -\tau^2$  from the pure-tether density). Hence  $f \equiv 0$ , yielding  $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$ , i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (36)$$

Integrating against  $\rho$ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (37)$$

Since  $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$ , rearranging gives (35).  $\square$

An immediate corollary follows.

**Corollary 8** (Closed-form  $T_2$ ).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (38)$$

This follows by substituting  $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2\sinh^2 \tau)$  and  $\rho(0) = \tau \coth \tau$  into the theorem.

The identity provides a closed-form evaluation of the allocation energy.

#### A.8 ALLOCATION STRENGTH AND A REFERENCE RULE

The Cosh family varies with the strength  $\tau$ . A reference used in the full-RoPE text experiments relates this strength to head width and training length:

$$\tau = c d_{\text{head}} / \sqrt{L_{\text{train}}}, \quad u_k = (k + \frac{1}{2})/K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (39)$$

The reference sets  $c = 1$  before training. Table 7 records the strength, grid, and setting procedure used in each experiment.

**Local scaling calculation.** The scaling argument uses four explicit modelling assumptions: (i) the pure-tether Cosh family is varied only through  $\tau$ ; (ii) full-RoPE MHA has  $d_{\text{rot}} = d_{\text{head}}$ ; (iii) the reference post-softmax distribution is diffuse,  $p_0 = 1/L$ ; and (iv) a fixed trade-off  $\lambda > 0$  compares a normalised channel-load stiffness with a specified phase-variance utility, with  $Q_1(L, b) > 0$ . Under those assumptions and at small  $\tau$ ,

$$S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 \frac{(1 - \rho_\tau(\phi))^2}{\rho_\tau(\phi)} d\phi = \frac{\tau^4}{45d_{\text{head}}} + O(\tau^6), \quad (40)$$

$$U(\tau, L) = \frac{d_{\text{head}}}{L} [Q_0(L, b) + \tau^2 Q_1(L, b) + O(\tau^4)], \quad (41)$$

where

$$\begin{aligned} Q_0(L, b) &= \int_0^1 q(Lb^{-\phi}) d\phi, \\ Q_1(L, b) &= \int_0^1 \left( \frac{(1-\phi)^2}{2} - \frac{1}{6} \right) q(Lb^{-\phi}) d\phi, \\ q(x) &= \frac{1}{2} + \frac{\sin(2x)}{4x} - \left( \frac{\sin x}{x} \right)^2. \end{aligned} \quad (42)$$

Stationarity of  $\mathcal{F}(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda U(\tau, L)$  gives the leading small- $\tau$  balance

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}, \quad c_{\text{loc}}(L, b, \lambda) = \sqrt{45\lambda Q_1(L, b)}. \quad (43)$$

This calculation yields the scaling factor  $d_{\text{head}}/\sqrt{L}$  with a local coefficient determined by  $Q_1(L, b)$  and  $\lambda$ . The experiments evaluate the exact Cosh quantiles at the protocol’s chosen strength.

**Reference and neighboring strengths.** The staged study contains 99 completed runs: the fixed  $c=1$  reference beats the midpoint-discretised Geo baseline under runner-defined weighted extrapolation NLL in 7/9 configuration means and 18/27 paired seeds. Its registered neighbours are finite multiplier arms at  $0.75\times$ ,  $1.25\times$ , or  $1.5\times$ ; the independent exact-range factorial tests  $0.75\times$ ,  $1.00\times$ , and  $1.25\times$ , and their order varies by configuration. The numeric owner is `data/curated/phase16_99run_manifest.csv`; the tracked reproduction entropoint is `scripts/core_text_phases/phase16_formula_optimality_sweep.py`. The reference and its neighboring strengths give a direct comparison of allocation intensity across the tested configurations.

#### A.9 DISCRETE-CHANNEL TRANSPORT GAP UNDER FINITE CHANNEL COUNTS

**Setup.** Inverse-CDF quantization replaces the continuous density  $\rho$  by an atomic measure. Let  $F(\phi) = \int_0^\phi \rho$ ,  $Q(u) = F^{-1}(u)$ , midpoint grid  $u_k = (k - \frac{1}{2})/K$ , and  $\phi_k = Q(u_k)$ . The empirical channel measure is  $\mu_K = K^{-1} \sum_{k=1}^K \delta_{\phi_k}$ . We compare the measures in Wasserstein distance and approximate the density with the associated quantile-cell histogram  $\rho_K$ .

**Transport bounds.** Assume  $0 < m \leq \rho \leq M$  and  $\|\rho'\|_\infty \leq B$ ; then  $Q$  is  $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (44)$$

Defining the quantile-cell histogram  $\rho_K(\phi) = 1/(K|I_k|)$  on  $I_k = [Q((k-1)/K), Q(k/K)]$ , the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (45)$$

The transport bounds couple each quantile cell to its midpoint; the histogram bound follows from the mean-value theorem and  $|I_k| \leq 1/(Km)$ .

**Application to EVQ-Cosh.** For  $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$ ,  $m_\tau = \tau/\sinh \tau$  and  $B_\tau = \tau^2$ . Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (46)$$

**Kernel-integrated errors.** For a smooth two-variable kernel  $K_{\text{sm}}$  with Lipschitz constant  $L_K$  in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km}. \quad (47)$$

The kernel bound applies the  $W_1$  estimate to each argument. The singular  $\delta$ -component of  $K_{\text{app}}$  is represented by its cell average.

**High-resolution distortion.** For midpoint scalar quantisation with a smooth weighted distortion  $w$ , the standard high-resolution (Bennett-integral) expansion (Gray & Neuhoﬀ, 1998) is

$$\mathcal{D}_K[\rho] = \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (48)$$

This expansion quantifies how finite exponent samples approximate a continuous allocation under a smooth distortion. For endpoint-anchored comparisons,  $K = 2$  leaves no interior entries and the geometric and Cosh grids coincide.

**Large- $\tau$  regime.** At large  $\tau$ ,  $m_\tau = \tau/\sinh \tau$  becomes small and the  $1/(Km_\tau)$  factor grows. The transport bounds are informative while this factor remains controlled.

## B EXPERIMENTAL DETAILS

### B.1 EXPERIMENT ORGANIZATION

This section gives the training, continuation, and video protocols. Apps. C and E detail the fixed-range and mature-model experiments.

### B.2 MODELS AND TRAINING SETTINGS

Table 6 collects the model settings and evaluation units. Scratch text runs use FineWeb-Edu (Penedo et al., 2024).

Table 8 lists the training hyperparameters shared across the supporting text experiments.

Within each text comparison, the architecture, optimizer, data order, and training loop are fixed, and the only intervention is the initialized inverse-frequency table. The scratch-text experiments use base  $b=500K$  unless stated otherwise.

Table 6: Evidence and protocol map. Every row retains its own intervention, experimental unit, and endpoint.

| Role                        | Model/regime and intervention                          | Unit, endpoint, and appendix                            |
|-----------------------------|--------------------------------------------------------|---------------------------------------------------------|
| Allocation identification   | 151.9M scratch; fixed support, 30 interiors            | 3 training seeds; tail NLL; App. C.1                    |
| Co-adaptation replication   | 151.9M frozen crossing; weights $\times$ runtime table | 2 training seeds; tail NLL; App. C.2                    |
| Shape/configuration breadth | 50.9M factorial; fixed support, 5 allocations          | 12 configs $\times$ 3 seeds; weighted OOD NLL; App. C.3 |
| Scarce rotary budget        | 432M MLA-style; Geo/EVQ-COSH, $K=16$                   | 3 training seeds; PPL; App. G.1                         |
| Full-parameter scale        | 750M/1.485B; continued/from-init Geo/EVQ-COSH          | one comparison each; PPL/retrieval; Apps. B.4/E.1       |
| Frozen allocation           | OLMo/Qwen; same support and amplitude                  | 20 rows/task; RULER; App. E.2                           |
| Zero-training deployment    | OLMo; frozen long table, gain, session route           | fixed checkpoint/rows; NLL/RULER/F1; App. E.2           |
| Adaptation/source use       | 1.485B/8B; matched LoRA/source block                   | one trained pair; QA/RULER/NLL; Apps. E.1/E.3           |
| Static index profile        | Qwen-0.5B; one table across lengths                    | 13 tasks, 20 rows/task/length; App. F.1                 |
| Profile placement           | Qwen/Gemma; direct gap versus normalized index         | 4 tasks, 80 rows/task; App. F.2                         |
| Intermediate-band shape     | OLMo/Qwen; matched BM/MrPro profiles                   | 6 tasks; Table 4; App. F.4                              |
| Natural-QA transfer         | OLMo; matched static BM/MrPro profiles                 | 778 inputs, 631 long; token F1; App. F.5                |
| Video modeling              | 129.6M video DiT; temporal Geo/EVQ-COSH                | 1 training seed; denoising MSE; App. B.6                |

Table 7: Cosh-family settings used in the reported experiments. Choosing  $\tau$  and a grid specifies a fixed table. Midpoint means  $u_k = (k + 1/2)/K$ ; standard means  $u_k = k/K$ ; anchoring fixes the sampled endpoints after evaluating the quantiles.

| Protocol           | $\tau$ | Grid               | Setting convention                                                       |
|--------------------|--------|--------------------|--------------------------------------------------------------------------|
| 151.9M fixed-range | 4      | Anchored midpoint  | Fixed $64/\sqrt{256}$ reference                                          |
| 50.9M factorial    | Varies | Anchored midpoint  | Three preassigned multiples of $d_{\text{head}}/\sqrt{L_{\text{train}}}$ |
| 50M crossing       | 2.83   | Midpoint           | Reported training-table setting                                          |
| 432M MLA           | 1.414  | Midpoint           | Pre-specified architecture-specific setting                              |
| 454M composition   | 1.5    | Midpoint           | Reported training recipe                                                 |
| 750M continuation  | 1.5    | Anchored inclusive | Fixed continuation recipe; $u_k = k/(K - 1)$                             |
| Llama-3-8B         | 1.414  | Midpoint           | Fixed adaptation recipe; approximately $128/\sqrt{8192}$                 |
| OLMo-2             | 2      | Standard           | Fixed $128/\sqrt{4096}$ reference                                        |
| 129.6M Video DiT   | 1.5    | Midpoint           | Best tested value in the video sweep                                     |

Table 8: Training hyperparameters for the supporting scratch text experiments.

| Parameter                 | Value                                                         |
|---------------------------|---------------------------------------------------------------|
| Optimizer                 | AdamW ( $\beta_1=0.9$ , $\beta_2=0.95$ , $\epsilon=10^{-8}$ ) |
| Learning rate             | $6 \times 10^{-4}$ (50M), $2 \times 10^{-4}$ (MLA/750M)       |
| LR schedule               | Cosine decay to $0.1 \times$ peak LR                          |
| Warmup                    | 2% of total steps (linear)                                    |
| Batch size (seqs)         | 32 (50M), 14 (750M: micro 7, accumulation 2), 6 (MLA)         |
| Weight decay              | 0.1                                                           |
| Gradient clipping         | max-norm 1.0                                                  |
| Dropout                   | 0.0                                                           |
| Sequence packing          | Contiguous, no cross-document masking                         |
| <i>Video DiT (129.6M)</i> |                                                               |
| Optimizer                 | AdamW ( $\beta_1=0.9$ , $\beta_2=0.95$ )                      |
| RoPE base                 | 10,000 (temporal axis)                                        |
| Activation                | GELU (tanh approx.)                                           |
| <i>MLA (432M)</i>         |                                                               |
| LR / warmup               | $2 \times 10^{-4}$ / 2% of steps                              |
| Training tokens           | 500M                                                          |

### B.3 COMPOSITION WITH A FIXED-INDEX RANGE OPERATOR

The 454.2M decoder has 24 layers, width 1024, 16 heads,  $d_{\text{head}} = 64$ , and base  $b = 500,000$ . Geo ( $\tau = 0$ ) and EVQ-COSH ( $\tau = 1.5$ ) use midpoint frequency grids and train from initialization for 100M tokens per arm at length 2048, using FineWeb-Edu with a 10% synthetic-passkey mixture and seeds 42, 123, 7. The same inference-time operator is then applied to both trained tables.

For  $k = 0, \dots, K - 1$ , define

$$\ell = \lfloor 0.20K \rfloor, \quad h = \lfloor 0.90K \rfloor, \quad t_k = \text{clip}\left(\frac{k - \ell}{h - \ell}, 0, 1\right), \quad r_k = t_k^2(3 - 2t_k).$$

The repository’s fixed-index, YaRN-style smooth-ramp operator installs

$$\mathcal{R}_s(\omega_k) = \frac{\omega_k}{s^{r_k} T(s)^{r_k/2}}, \quad T(s) = 1 + 0.07 \log_2 s. \quad (49)$$

Here  $K = 32$  and  $s = 8$ , giving  $(\ell, h, T) = (6, 28, 1.21)$ . The temperature factor is folded into the frequency divisor; cosine/sine amplitude stays at  $g = 1$ . Equation (49) specifies the operator used in this experiment, separately from the official-equation YaRN comparisons in §6. Table 9 reports the three-seed comparison.

Table 9: Matched fixed-ramp composition on the 454M three-seed experiment.  $\mathcal{R}_8$  is defined in Eq. (49). PPL uses full-sequence scoring; PK is teacher-forced NLL-gap retrieval. Entries are three-seed means; PK@8K also gives the standard deviation.

| Table                     | PK@8K         | PK@12K     | PK@16K     | PPL@8K      | PPL@16K      |
|---------------------------|---------------|------------|------------|-------------|--------------|
| Geo                       | 41±5%         | 57%        | 51%        | 161.9       | 253.2        |
| Geo+ $\mathcal{R}_8$      | 61±3%         | 59%        | 51%        | 82.9        | 157.7        |
| EVQ-COSH                  | 53±8%         | 63%        | 50%        | 150.3       | 229.5        |
| EVQ-COSH+ $\mathcal{R}_8$ | <b>100±0%</b> | <b>79%</b> | <b>68%</b> | <b>70.9</b> | <b>107.5</b> |

### B.4 LARGER-SCALE SUPPORTING EVIDENCE

The 750M ‘continue@4K’ experiment starts both arms from the same 2K Geo checkpoint and applies a 500M-token continuation with either the Geo or EVQ-COSH table (seed 42). At 16K, PPL is 45.1 versus 24.4; on 40 passkey trials at 8K, strict autoregressive exact match is 0% versus 77.5%. The EVQ-COSH continuation uses  $\tau = 1.5$ ,  $K = 32$ , and the endpoint-inclusive quantiles  $u_k = k/(K - 1)$ . Writing  $q_\tau$  for the inverse CDF in Eq. (8), the installed frequencies are

$$\omega_k = \exp\left[-\frac{K - 1}{K} \log b q_{1.5}\left(\frac{k}{K - 1}\right)\right], \quad b = 500,000.$$

This retains the Geo endpoints. Both arms are full-parameter continuations from the shared checkpoint.

Table 10: 750M continued-pretraining check (seed 42; 2K→4K; 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials per length. The pooled AR row combines 2K, 4K and 8K (120 trials: Geo 80/120, EVQ-COSH 111/120).

| Metric                            | Geo    | EVQ-COSH     |
|-----------------------------------|--------|--------------|
| PPL@2K                            | 25.9   | 26.2         |
| PPL@4K                            | 22.0   | 22.3         |
| PPL@8K                            | 23.4   | <b>19.6</b>  |
| PPL@16K                           | 45.1   | <b>24.4</b>  |
| NLL-gap retrieval @8K             | 100%   | 100%         |
| Strict AR exact @8K               | 0%     | <b>77.5%</b> |
| Pooled strict AR exact (2K/4K/8K) | 66.67% | <b>92.5%</b> |

## B.5 MODALITY-DEPENDENT $\tau$ CORRECTION FOR VIDEO DiT

The text-style reference used for this sweep gives  $\tau = K_t / \sqrt{T} = 2.83$  for  $K_t=16$  and  $T=32$ ; the best tested Video DiT point is  $\tau=1.5$  ( $0.53\times$ ), which selects the supporting video configuration.

## B.6 CROSS-MODAL EVIDENCE: VIDEO DiT WITH BIDIRECTIONAL ATTENTION

VideoRoPE and RIFLEx make temporal-frequency allocation explicit in video models (Wei et al., 2025; Zhao et al., 2025). We test the same allocation principle in a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015). Training uses 32 frames; evaluation uses 128 frames ( $4\times$ ) under the same fixed temporal range transform in both arms. The seed-42 Geo and EVQ-COSH ( $\tau=1.5$ ) branches are trained for 15,000 steps in the same head-to-head run. The denoising endpoint uses noise level  $t=0.5$  and the same 256 evaluation videos for both arms.

Table 11 reports the matched comparison. EVQ-COSH reduces denoising MSE by 21% on training frames, 16% over all extrapolated frames, and 35% on far extrapolated frames under bidirectional 3D RoPE.

Table 11: DiT head-to-head evaluation (129.6M, seed 42, 15,000 steps, 256 evaluation videos, bidirectional attention, 3D RoPE). Values are denoising MSE at  $t=0.5$ .

| Seed | Metric     | Geo ( $\tau=0$ ) | EVQ-COSH ( $\tau=1.5$ ) | $\Delta$ |
|------|------------|------------------|-------------------------|----------|
| 42   | Train MSE  | 0.00911          | <b>0.00720</b>          | −21%     |
| 42   | All extrap | 0.00724          | <b>0.00606</b>          | −16%     |
| 42   | Far extrap | 0.00989          | <b>0.00639</b>          | −35%     |

**Low-phase-channel diagnostic.** At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. This analytic count identifies the low-phase regime tested by the matched behavioural comparison above.

## B.7 WHAT THE SCHEDULE CONCRETELY DOES

At  $\tau=4$ ,  $b=5 \times 10^5$ , the warp moves budget toward short inverse-frequency scales: the continuous quantile  $u=0.5$  moves from  $1/\omega=707$  to 9.69, and  $u=0.9$  from 134,609 to 1,720. The deployed midpoint grid therefore changes sampled extrema and span at finite  $K$ ; the exact-range experiments anchor those quantities explicitly. At  $\tau=2$ ,  $\rho_\tau(0.1)=1.71$  and  $\rho_\tau(1)=0.55$ , while the band  $1/\omega=2048$  has relative density 0.76. Thus the finite grid thins most strongly at scales much slower than the training window while retaining more of the training-scale band. The exact-range experiments measure the corresponding trained allocation effect, and Table 5 separately measures incompatibility after a frozen post-hoc swap.

## C CONTROLLED ALLOCATION EXPERIMENTS

### C.1 EXACT-RANGE CONTROL AT 151.9M

**Paired controls.** The two trained tables have identical frequency endpoints. Within each seed pair the arms share: model architecture (151,898,880 parameters); trainable initialisation; training length 256; token order; optimiser, LR schedule, global batch; within-pair micro batch and accumulation; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the  $K-2=30$  interior frequencies.

The optimiser is fused AdamW with  $\beta_1=0.9$ ,  $\beta_2=0.95$ , weight decay 0.01, and gradient clipping at 1.0. Peak/minimum learning rates are  $6 \times 10^{-4}/6 \times 10^{-5}$  with 762 warmup steps and cosine decay. The global batch is 256 sequences of length 256 for every arm. Seed 42 uses micro batch 64 with four accumulation steps; seeds 137 and 256 use micro batch 128 with two accumulation steps. Each

paired comparison therefore has identical execution geometry, optimiser steps, and counted-token budget; the aggregate uses training seed as its statistical unit under this shared scientific protocol.

**Model and data.** The model has 12 layers, hidden size 768, 12 heads of dimension 64, MLP width 3,072, and vocabulary size 50,304. Training uses the unshuffled FineWeb-Edu (Penedo et al., 2024) sample/10BT/000\_00000.parquet shard at pinned revision 87f09149ef4734204d70ed1d046ddc9ca3f2b8f9; validation uses the disjoint sample/10BT/004\_00000.parquet shard. Both are tokenised with the GPT-NeoX-20B tokenizer (Black et al., 2022) at revision c292233c833e336628618a88a648727eb3dff0a7.

**Arms.** *Baseline:* FMRoPE at fixed training support, implementing the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026b) at training base 256. *Intervention:* anchored EVQ-COSH at  $\tau=4$ . The anchored and deployed midpoint EVQ-COSH tables are different support embeddings of the *same normalised shape*: both use midpoint Cosh quantiles  $q_k$ , and the diagnostic removes only their affine endpoint degrees of freedom. This normalisation is what makes the exact-range control possible.

Writing  $\Phi_\tau$  for the inverse CDF in (8), let  $q_k = \Phi_\tau((k + \frac{1}{2})/K)$ ,  $s_k = (q_k - q_0)/(q_{K-1} - q_0)$ , and  $R = ((K-1)/K) \log B$ . The intervention uses

$$\omega_k^{\text{anchor}} = \exp(-Rs_k), \quad \omega_k^{\text{FMR}} = \exp\left(-R \frac{k}{K-1}\right). \quad (50)$$

Hence  $k=0, K-1$  match exactly and only the  $K-2$  interior entries differ.

**Evaluation.** Two conditions. *fixed:* each checkpoint retains its training range. *target-matched:* the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is  $\exp(\text{mean NLL})$ . The anchors are common end positions selected with Python’s `random.Random(20260723)`: the eligible validation stream is divided into 32 equal-width bins, with one end position drawn per bin and a 2,048-token guard before the next bin. At anchor  $e$ , length  $L$  uses validation tokens  $[e-L, e]$  and scores the final 128 targets  $[e-128, e]$ . Longer contexts therefore add earlier text around the same target tokens; the longest windows at distinct anchors are disjoint.

**Exact range transformation.**

$$R(L) = \frac{K-1}{K} \log L, \quad \omega_k^{\text{target}}(L) = \exp[-R(L)z_k].$$

Here  $z_k = k/(K-1)$  for FMRoPE and the anchored Cosh shape for its paired arm. The fixed condition uses  $L = 256$  at every evaluation length; the target-matched condition substitutes the declared evaluation length. Both conditions use amplitude one.

**Three-seed contrasts.** Negative values favour anchored EVQ-COSH. The aggregation unit is the training seed; evaluation anchors remain paired observations within a seed.

Table 12: Exact-range anchored EVQ-COSH minus FMRoPE tail-NLL contrasts. Mean $\pm$ SD is across three independent training seeds.

| Length | Fixed training range |          |          |                    | Target-matched range |          |          |                    |
|--------|----------------------|----------|----------|--------------------|----------------------|----------|----------|--------------------|
|        | Seed 42              | Seed 137 | Seed 256 | Mean $\pm$ SD      | Seed 42              | Seed 137 | Seed 256 | Mean $\pm$ SD      |
| 256    | +0.033               | +0.029   | +0.017   | +0.026 $\pm$ 0.008 | +0.033               | +0.029   | +0.017   | +0.026 $\pm$ 0.008 |
| 512    | -0.478               | -0.271   | -0.094   | -0.281 $\pm$ 0.192 | +0.061               | +0.053   | +0.067   | +0.060 $\pm$ 0.007 |
| 1,024  | -0.205               | -0.191   | -0.132   | -0.176 $\pm$ 0.039 | +0.182               | +0.156   | +0.344   | +0.227 $\pm$ 0.102 |
| 2,048  | -0.113               | -0.181   | -0.143   | -0.146 $\pm$ 0.034 | +0.279               | +0.399   | +0.701   | +0.460 $\pm$ 0.218 |

Across all three seeds, fixed training range favors Cosh at every extended length, whereas target-matched range favors FMRoPE. The paired design thus shows both an interior-allocation effect and a change in preferred allocation when the same trained weights are evaluated under a retargeted range.

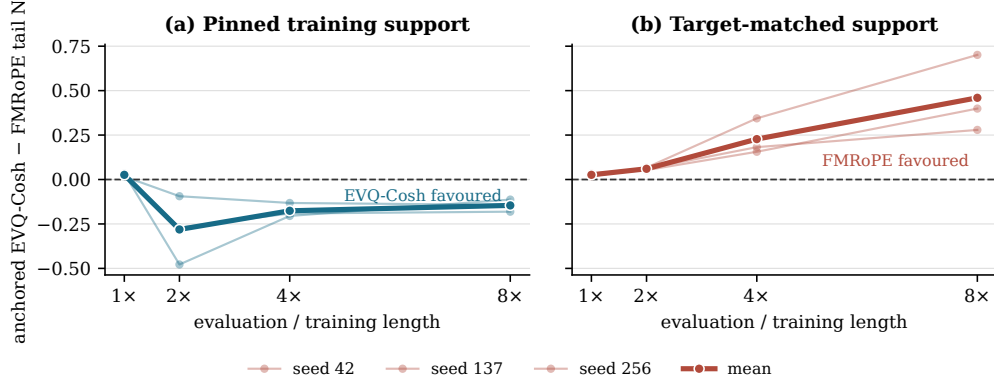


Figure 7: **Support and interior allocation are distinct design coordinates.** Each thin line is one training seed and the heavy line is the mean. At pinned training support, all three seeds favour anchored EVQ-COSH at every OOD length (a). Retargeting each arm’s support reverses all three OOD comparisons (b). The fixed-support panel isolates the allocation effect; the tested target-matched policy reverses the ordering of the two tested policies.

## C.2 WEIGHTS-BY-TABLE CO-ADAPTATION AT 151.9M

The three-seed experiment above identifies the training-time effect of  $z$ . A separate frozen crossing asks whether the learned weights remain compatible with the coordinate system on which they were trained. Seeds 137 and 256 provide paired FMRoPE and anchored-EVQ-COSH checkpoints whose training tables share exact endpoints. For each frozen checkpoint, we install the factor-four runtime table derived from either training coordinate, keep the checkpoint, amplitude, 32 FineWeb-Edu validation anchors, and final-128-token NLL endpoint fixed, and evaluate at 512 and 1,024 tokens.

For each training table  $\Omega$ , the derived table uses the following label-free residual construction. Set  $L_0 = 256$ ,  $\Delta_j = 2j$  for  $j = 0, \dots, 127$ , and  $p_j = (2L_0 - 4j - 1)/(L_0(L_0 + 1)/2)$ , the grouped causal-separation weights. Let  $X_k$  have columns  $\sqrt{p_j} \cos(\omega_k \Delta_j)$  and  $\sqrt{p_j} \sin(\omega_k \Delta_j)$ , and let  $X_{-k}$  collect all other pairs. Define

$$U_k = \frac{\|X_k - X_{-k} X_{-k}^\dagger X_k\|_F^2}{\|X_k\|_F^2}, \quad \bar{U}_k = \frac{U_k - \min_i U_i}{\max_i U_i - \min_i U_i}, \quad w_k = (1 - \bar{U}_k)^2.$$

The pseudoinverse uses relative singular-value cutoff  $10^{-10}$ ; the zero-span convention is  $\bar{U}_k = 0$ . The frequency blend is  $\omega'_k = \omega_k(1 - 3w_k/4)$ , cast to FP32 with its endpoints set exactly to  $\omega'_0 = \omega_0$  and  $\omega'_{K-1} = \omega_{K-1}/4$ . Both derived tables use  $g = 1 + 0.1 \log 4$  and remain unchanged across the 512- and 1,024-token evaluations.

Table 13: Two-training-seed mean tail NLL at 1,024 tokens for the 151.9M weights-by-table crossing (lower is better). Each row freezes one trained weight set; each column installs one separately derived runtime table.

| Frozen weights            | FMRoPE-derived table | Anchored-EVQ-COSH-derived table |
|---------------------------|----------------------|---------------------------------|
| FMRoPE-trained            | <b>3.426</b>         | 5.776                           |
| Anchored-EVQ-COSH-trained | 4.455                | <b>3.479</b>                    |

With  $I = [L(W_F, T_C) - L(W_F, T_F)] - [L(W_C, T_C) - L(W_C, T_F)]$ , the interaction is 3.400/3.251 NLL for seeds 137/256 at 1,024 tokens and 2.663/2.564 at 512. Training seed is the replication unit; the 32 anchors are paired observations within a seed. Both seeds exhibit the same preference for the runtime table derived from their training allocation.

Table 14: **Exact-range factorial** (50.9M; 2 bases  $\times$  2 training lengths  $\times$  3 head dimensions = 12 configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over  $2 \times /4 \times /8 \times$ , paired within configuration and seed. All arms were pre-specified. CIs are 10,000 configuration-level bootstraps;  $p$  is a two-sided exact sign-flip over all  $2^{12}$  assignments, unadjusted across contrasts, so the claim rests on the cross-configuration direction (App. C).

| Contrast                                                             | $\Delta$ NLL     | 95% CI           | $\Delta < 0$ configs | $p$   |
|----------------------------------------------------------------------|------------------|------------------|----------------------|-------|
| EVQ-COSH $0.75 \times$ – Geo                                         | −0.00912         | [−0.018, −0.001] | 8/12                 | 0.082 |
| EVQ-COSH rule – Geo                                                  | −0.00988         | [−0.021, 0.002]  | 7/12                 | 0.125 |
| EVQ-COSH $1.25 \times$ – Geo                                         | − <b>0.01210</b> | [−0.021, −0.003] | <b>10/12</b>         | 0.027 |
| Matched exponential – Geo                                            | −0.01062         | [−0.021, −0.001] | 9/12                 | 0.071 |
| <i>The matched exponential reproduces the non-uniform direction:</i> |                  |                  |                      |       |
| EVQ-COSH rule – exponential                                          | +0.00074         | [−0.006, 0.008]  | 7/12                 | 0.836 |

### C.3 EXACT-RANGE FACTORIAL AT 50.9M

**Design.** The 50.9M decoder has six layers, width 512, MLP width 2,048, and vocabulary size 50,304. The structural grid is RoPE base 500K/1M  $\times$  training length 256/1,024  $\times$  4/8/16 heads ( $d_{\text{head}}=128/64/32$ ), giving 12 configurations. Seeds are 42/137/256. Every arm trains for 8,388,608 tokens in 128 optimizer steps (65,536 tokens per step) on the same deterministically repeated WikiText-2 raw stream with the same 1% supervised passkey mix; execution is float32 on Apple M4 Max MPS. The schedule arms, including the  $1.25 \times$  multiplier, were specified before any result was inspected: Geo, anchored EVQ-COSH at  $0.75 \times$ ,  $1.00 \times$  (the formula point),  $1.25 \times$ , and a deformation-matched exponential. Main factorial 180/180 completed and the boundary- $\tau$  follow-up 12/12 completed, covering the pre-specified matrix.

The exponential control uses  $z_\gamma(t) = (e^{\gamma t} - 1)/(e^\gamma - 1)$  on  $t \in [0, 1]$ . Its  $\gamma$  is chosen without task results so that the RMS node displacement from the uniform grid matches the formula-Cosh displacement, while the sampled endpoints and log-span remain identical across arms.

**Metric.** Weighted OOD NLL over  $2L/4L/8L$ ,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are  $\exp(\text{mean NLL})$ .

**Absolute values.** Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); EVQ-COSH  $0.75 \times$  6.202858 (494.16); EVQ-COSH  $1.00 \times$  6.202094 (493.78); EVQ-COSH  $1.25 \times$  6.199873 (492.69); matched exponential 6.201354 (493.42).

**Boundary- $\tau$  arms.** At formula  $\tau=1$  the  $1.5 \times$  arm is better; at formula  $\tau=8$  the  $0.75 \times$  arm is. The improving direction points toward the intermediate strengths at both tested extremes.

The factorial varies architecture, base, and training length on a repeated WikiText-2 stream (Merity et al., 2017). Its configuration-level contrasts complement the longer FineWeb-Edu training comparison in §C.1.

## D LEARNED-FREQUENCY COMPARATOR

The 32-parameter arm learns one inverse frequency per rotary pair, shared across layers and heads. This is the learned-table parameterization also studied by Karypis et al. (2026). Its positional-parameter learning-rate sweep uses  $10 \times$  and  $100 \times$  the base rate; the stronger  $100 \times$  setting gives 455.3 PPL. The learned table and the fixed Cosh allocation both improve on Geo in this protocol.

## E MATURE-SCALE PROTOCOLS

Table 15: Learned-frequency comparator (125M,  $L_{\text{train}}=128$ , FineWeb-Edu,  $128 \rightarrow 8K$ ). Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- $\tau$  is 3-seed (42/137/256). The learned-inverse-frequency arm has one parameter per pair, shared across layers and heads.

| Method (128 $\rightarrow$ 8K)                 | Extra params | PPL@128                           | PPL@8K           | $\Delta$ vs Geo |
|-----------------------------------------------|--------------|-----------------------------------|------------------|-----------------|
| Geo                                           | 0            | 184.9                             | 513.7            | —               |
| Learnable $\tau$                              | 1            | <b>181.2 <math>\pm</math> 1.3</b> | 437.9 $\pm$ 12.2 | -14.8%          |
| Learned inv-freq (100 $\times$ positional LR) | 32           | 183.6                             | 455.3            | -11.4%          |
| same, 10 $\times$ positional LR               | 32           | —                                 | 477.7            | -7.0%           |
| EVQ-COSH                                      | 0            | 182.0                             | <b>333.7</b>     | <b>-35.0%</b>   |

Table 16: **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm; evaluation uses 20 examples per family and length. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on; LLaMA uses matched Q/K/V/O LoRA. Training and evaluation rows are disjoint and share the 13 generator families, measuring task-family-adapted length transfer.

| Model / protocol                     | Arm / metric                    | 1 $\times$   | 2 $\times$   | 4 $\times$  |
|--------------------------------------|---------------------------------|--------------|--------------|-------------|
| OLMo-2 1.485B, full Q/K/V/O (4K cap) | Native, official-scorer macro   | <b>82.16</b> | 0.08         | 0           |
|                                      | EVQ-COSH, official-scorer macro | 37.51        | <b>21.29</b> | <b>6.13</b> |
| OLMo-2 1.485B, Q/K-only (4K cap)     | Native, official-scorer macro   | <b>72.19</b> | 2.02         | 0.38        |
|                                      | EVQ-COSH, official-scorer macro | 42.44        | <b>31.63</b> | <b>5.03</b> |
| LLaMA-3-8B, Q/K/V/O (8K cap)         | Native, official-scorer macro   | <b>94.44</b> | 0.295        | —           |
|                                      | EVQ-COSH, official-scorer macro | 77.60        | <b>14.03</b> | —           |
|                                      | Native, normalized exact        | 17.69        | 0            | —           |
|                                      | EVQ-COSH, normalized exact      | <b>21.54</b> | <b>1.54</b>  | —           |

### E.1 OLMo-2 1.485B

All OLMo-2 EVQ-COSH arms use the standard  $u_k=k/K$  grid with  $\tau=2$  and base 500,000; the frequency tensor is fixed throughout each continuation.

**Matched routing conversion.** We use OLMo-2-0425-1B-Instruct (Team OLMo et al., 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-COSH share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical tokens, with explicit position IDs providing target-range relative phases. On 8K niah\_single\_1, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set.

**Complete numeric answers after matched routing continuation.** A separate follow-up to the routing parents applies the same 100-step query-gap continuation and 32-step answer-plus-EOS continuation to Native and EVQ-COSH (training seed 20260728). Every backward pass uses at most 4096 physical tokens; explicit position IDs expose target-range phases, with maximum observed position ID 16,257. Evaluation uses real contiguous prompts from the same numeric NIAH family and greedy decoding. The endpoint requires the complete expected answer string followed by observed terminal EOS, measured from generated token IDs.

Table 17: Complete-answer-string plus terminal-EOS exact match after the matched routing follow-up, 100 inputs per length and one trained pair.

| Length | Native | EVQ-COSH |
|--------|--------|----------|
| 4K     | 95/100 | 100/100  |
| 8K     | 18/100 | 98/100   |
| 16K    | 0/100  | 60/100   |

All final generations in both arms terminate with EOS. On the EVQ-COSH subsets whose source-query gap exceeds the original training range, exact match is 48/50 at 8K and 31/66 at 16K. These outputs belong to the additional continuation above, separately from the earlier first-number retrieval comparison.

**Selective Q/K phase adaptation.** To test a less intrusive mature-checkpoint adaptation, we start from the protocol-matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O LoRA tensor within each arm (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Each arm trains 8,388,608 Q/K parameters while freezing the 8,388,608 inherited V/O parameters, with global batch 8, learning rate  $5 \times 10^{-5}$ , and 20 warmup steps. The QA and RULER generators use seeds 20260728 and 20260729, respectively; the bundled `Q/K metrics.json` records the full protocol. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length (Ho et al., 2020), Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K; exact match is 22.0/21.5%, 0/17.5%, and 0/4.0%, respectively. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length, forming a held-out 2Wiki length-transfer evaluation. The RULER adapters are independent of the QA adapters: Table 16 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families. Evaluation uses 20 examples per family and length. A paired family-stratified bootstrap over the tracked Q/K-only per-example predictions gives EVQ-COSH-minus-Native official-scorer macro differences of  $-29.75$  points at 4K (95% CI  $[-34.42, -24.85]$ ),  $+29.61$  at 8K ( $[26.54, 32.74]$ ), and  $+4.65$  at 16K ( $[3.17, 6.38]$ ). These intervals quantify evaluation-row variation for the trained pair.

**From initialisation.** Starting from the public step-0 initialisation, the same-recipe Geo/EVQ-COSH comparison gives PPL 177.99/191.36, 161.19/167.45, 163.88/156.87, and 182.73/159.64 at 2/4/8/16K under a matched 2.097B-token budget. The EVQ-COSH-minus-Geo NLL is  $+0.0724/+0.0381/-0.0437/-0.1351$  at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K.

The EVQ-COSH branch uses a Hugging Face single-GPU loop; the released Geo checkpoint was produced by AI2’s distributed OLMo trainer. Both start from the same public step-0 initialisation and match the pinned recipe, reconstructed seed-6198 data-order prefix, 512-sequence global batch, 4K context, 1,000 optimizer steps, and 2,097,152,000 counted input tokens. This comparison includes the frequency-table change and the two trainer implementations.

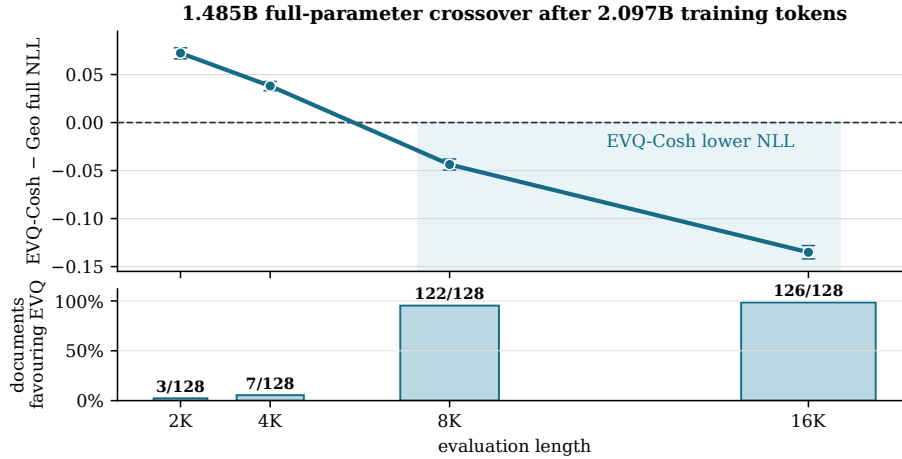


Figure 8: **Full-parameter scale evidence crosses beyond the training window.** The 1.485B Geo and EVQ-COSH arms start from the same public initialisation and follow the same scientific recipe for 2.097B tokens. Error bars are paired bootstrap intervals over the same 128 held-out documents, conditioned on this trained pair; the lower panel gives the document count favouring EVQ-COSH at each length.

## E.2 FROZEN FIXED-SUPPORT ALLOCATION

The mature-checkpoint intervention in Table 2 holds both models frozen. The three matched-support arms contain 64 rotary pairs and share the Native fast endpoint and the Native slow endpoint divided by four. The reference rows use the original Native table and the Transformers YaRN frequency map. The three matched-support arms also share the factor-four attention amplitude  $1 + 0.1 \ln 4$ , evaluation rows, greedy decoder, precision, and hardware. The uniform-allocation arm is log-linear between the endpoints. The coarse control is a label-free, model-relative projection of the frozen derived table: among the registered discrete linear ramps, it minimises displacement-space MSE, using transition pairs 20–22 for OLMo and 28–31 for Qwen. The remaining arm installs the frozen derived interior allocation. For that arm,  $\tilde{U}_k$  is the normalised residual-energy fraction left after the full sin/cos pair is projected onto all other pairs over Native separations weighted by their causal pair count,  $p(\Delta) \propto L - \Delta$ ; it sets  $w_k = (1 - \tilde{U}_k)^2$  in the frequency blend of Eq. (10), so less unique pairs move farther.

Table 18: **Exact frozen mature-checkpoint RULER macro (%)**. These are the values plotted in Table 2. Reference rows are separated from the fixed-support block; only the lower block changes interior  $z$  while holding the stated checkpoint protocol fixed.

| Installed table                      | OLMo unseen-nine, 16K | Qwen core-four, 64K |
|--------------------------------------|-----------------------|---------------------|
| Native                               | 0.00                  | 54.50               |
| Official Transformers YaRN ( $s=4$ ) | 7.94                  | 60.25               |
| Uniform allocation $z$               | 0.56                  | 57.75               |
| Coarse label-free $z$                | 61.04                 | 64.00               |
| Derived allocation $z$               | 60.47                 | 66.50               |

Method selection used the core-four tasks NIAH single-1, NIAH multikey-2/3, and variable tracking. After the table, amplitude, split rule, checkpoint, and metric were frozen, the remaining nine RULER families—NIAH single-2/3, multikey-1, multivalue, multiquery, CWE, FWE, and QA-1/2—formed the OLMo confirmation set at 16K, with 20 rows per task. Qwen2.5-1.5B-Instruct (Yang et al., 2024) uses the four development RULER tasks at 64K, also with 20 rows per task; it evaluates a factor-four table at twice the Native context length. Row resampling conditions on each fixed checkpoint and task set. Both reallocated tables produce the large OLMo recovery. The derived-minus-uniform-allocation intervals are  $[54.88, 64.80]$  points for OLMo and  $[0.25, 17.50]$  points for Qwen; the paired

derived-minus-coarse-control intervals contain zero on both models. The control therefore attributes the effect to fixed-support reallocation while leaving profile-specific ordering unresolved.

**Fresh-distribution allocation and routing controls.** On a disjoint 512-document FineWeb-Edu holdout from a shard absent from the calibration data, the session policy is exactly Native on all 512/512 4K rows. Table 19 separates its routing contrast from the fixed-support allocation contrasts. For the latter, support, attention gain, Native/long route, checkpoint, rows, and runtime are fixed; only interior  $z$  changes. The coarse ramp again matches the detailed derived profile, whereas the same-support geometric allocation is slightly better at 8K and fails on all 512 rows at 16K.

Table 19: Fresh FineWeb-Edu holdout-512 final-1,024-token NLL contrasts. The first row evaluates the complete session policy; the lower rows compare long allocations at fixed support, gain, and route. Negative favours the first named method.

| Contrast                  | 8K      | 16K     |
|---------------------------|---------|---------|
| Session-s4 — Native       | −4.0953 | −4.4393 |
| Geometric — derived $z$   | −0.0288 | +4.4560 |
| Coarse ramp — derived $z$ | +0.0005 | +0.0006 |

The OLMo deployment policy selects the exact Native table when observed prefill plus generation reserve fits the Native window; otherwise one frozen factor-four table and amplitude are selected before prefill and retained for the KV-cache lifetime. It adds no learned parameters or training tokens. Statically forcing the same long profile at 4K regresses NLL from 2.7538 to 2.8774; routing, rather than the long allocation, is therefore the identified in-window retention mechanism.

Table 20: Compact zero-training deployment endpoints on the frozen OLMo checkpoint, using the specified table, amplitude, and session routing.

| Endpoint                            | Official YaRN ( $s=4$ ) | Zero-training policy |
|-------------------------------------|-------------------------|----------------------|
| PG-19 tail NLL, 8K/16K              | 3.4404/3.7955           | <b>3.1060/3.0977</b> |
| RULER-13 macro, 8K/16K (%)          | 23.82/5.88              | <b>67.72/54.40</b>   |
| Qasper token F1, 16K ( $n=200$ , %) | 18.03                   | <b>24.57</b>         |
| 2Wiki token F1, 16K ( $n=200$ , %)  | 25.69                   | <b>26.66</b>         |

Applying the long profile to every 2Wiki row scores 27.74%; the routed policy pays that small difference to preserve the exact Native short path. Across the six-task natural-context matrix, routed/YaRN macro is 28.58/21.27% at  $2\times$  and 24.97/25.59% at  $4\times$ . The QA arms use identical rows, and every post-chat input plus generation reserve fits within 16K. Table 2 gives the corresponding allocation-only comparison.

### E.3 LLaMA-3-8B-INSTRUCT

The LLaMA-3-8B-Instruct checkpoint (Grattafiori et al., 2024) is frozen apart from the stated adapters. The EVQ-COSH arm in both LLaMA protocols uses midpoint quantisation with  $\tau=1.414$ ; Native retains the pretrained model’s original frequency table.

**Probability and causal source use.** The matched seed-42 protocol uses 300 optimizer steps of BF16 rank-64, alpha-128 LoRA on Q/K/V/O, physical 8,192-token sequences, and learning rate  $10^{-4}$ . Arms share model and tokenizer bytes, frozen training rows and order, seed, objective, optimizer, scheduler, batch/accumulation, LoRA capacity, checkpointing, compile mode, and evaluator. Native uses the pretrained endpoint grid, and Cosh uses midpoint quantiles; the same adaptation recipe is applied to these two frequency tables. The frozen adaptation tensor is drawn from LongAlpaca-12k (Chen et al., 2023b): 7,476 training and 152 validation rows after length filtering, with effective batch 8 (microbatch 2, accumulation 4).

**Temporal holdout and aggregation.** Evaluation uses arXiv titles/abstracts, Federal Register titles/abstracts, and Stack Overflow question bodies selected by publication or creation timestamps

between January 1 and July 12, 2026. These are separate from the LongAlpaca adaptation corpus. Collection was frozen before evaluation; each domain has eight disjoint-document 32,768-token packs formed by concatenation with EOS separators. Positions and causal attention continue across document boundaries. The 8K and 16K conditions are exact prefixes of the same 32K packs. All ordinary text targets in each prefix are scored, excluding document-first tokens and inserted EOS separators. For domain  $d$ , let  $\ell_d(L)$  and  $n_d(L)$  be the summed token loss and valid-token count over its eight packs. The reported aggregate is

$$\text{NLL}(L) = \frac{1}{3} \sum_{d=1}^3 \frac{\ell_d(L)}{n_d(L)}, \quad \text{PPL}(L) = \exp(\text{NLL}(L)).$$

Table 21 gives token counts and the distribution of the 24 paired pack-level NLL differences. The accompanying `figs/llama_temporal_summary.json` retains each pack’s loss sum and count, its domain identity, and the frozen collection hash.

Table 21: Llama-3-8B temporal holdout: scored tokens and the 24 paired pack NLL differences (EVQ-COSH minus Native). Each length uses the same packs.

| Prefix | Valid tokens/arm | Valid tokens/pack | Pack $\Delta\text{NLL}$ : min / median / max |
|--------|------------------|-------------------|----------------------------------------------|
| 8K     | 194,794          | 8,051–8,183       | +0.193/ + 0.432/ + 0.613                     |
| 16K    | 389,672          | 16,111–16,353     | −2.069/ − 1.436/ − 1.278                     |
| 32K    | 779,400          | 32,243–32,679     | −2.319/ − 2.074/ − 1.754                     |

The domain-macro EVQ-COSH-minus-Native NLL is +0.390/ − 1.510/ − 2.048 at 8/16/32K.

**Position-preserving source intervention.** On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Blocking answer-side decode attention to the remote gold block in every head preserves all prompt tokens, original rotary KV indices, and the 128-token block geometry. NLL pools the negative log probabilities of the 30 answer tokens (3 per case); deletion changes this mean by −0.0095 for Native and +1.5055 for EVQ-COSH. The EVQ-COSH change is positive in all ten cases. Case-level hit@16 ranges are 3.12–34.38% for Native and 62.50–68.75% for EVQ-COSH. These are matched teacher-forced probability and position-preserving causal-routing measurements; autoregressive task accuracy is evaluated separately. Descriptive Native/EVQ-COSH PPL is 6.817/10.068, 108.958/24.068, and 991.475/127.911 at those lengths.

**RULER-family continuation.** A separate seed-20420726 continuation uses 516 steps, rank-64/alpha-128 Q/K/V/O LoRA, global batch 8 (micro batch 2, accumulation 4), BF16 and Flash-only attention. Its 1,376 physical-8K training rows comprise 96 rows for each of 13 RULER families plus 128 natural-replay rows; it processes 33,816,576 input tokens. Training and evaluation rows have zero exact overlap, and evaluation uses 20 examples per family and length. At 16K, Native/EVQ-COSH official-scorer macro is 0.295/14.03% and normalized exact is 0/1.54%. At 8K, Table 16 reports official scorer macro (94.44/77.60%) and normalized exact (17.69/21.54%).

## F FROZEN EXPONENT-ADJUSTMENT PROTOCOLS

This section specifies the static profiles and evaluation panels used in §6. An amplitude  $g$  multiplies both the cosine and sine tables, so its contribution to a rotated Q/K logit is  $g^2$ . Every table is installed before prefill and retained for the entire request. The accompanying `figs/recorded_runtime_identities.json` records checkpoint, tokenizer, and evaluator identities. The index/placement studies use RULER commit `c3f5e3b4f87f97e048793bb510a3a6b19a46bf3a`.

## F.1 NORMALIZED-INDEX PROFILE

For a native table with  $K$  pairs, base  $b$ , and reference length  $L_{\text{ref}}$ , the normalized-index construction evaluates the profile on a 64-pair reference grid with the same  $b, L_{\text{ref}}$ :

$$\bar{\omega}_j = b^{-j/64}, \quad c_{64} = (1 - b^{-1/64})^{-1}, \quad \xi_j = \log \frac{L_{\text{ref}} \bar{\omega}_j}{2\pi c_{64}}, \quad (51)$$

$$\bar{m}_j = \text{clip} \left( \frac{\xi_H - \xi_j}{\xi_H - \xi_L}, 0, 1 \right), \quad \xi_H = 0.7382780681078285, \quad \xi_L = 0.366403835112904. \quad (52)$$

These two constants were fitted to the existing OLMo displacement profile and then frozen. Linear interpolation of  $(j/63, \bar{m}_j)$  at  $k/(K-1)$  gives  $m_k$  on the target grid. The installed table is  $\omega'_k = \omega_k^N s^{-m_k}$ .

The Qwen-2.5-0.5B-Instruct experiment uses its native  $K = 32$  pairs,  $b = 10^6$ , and original context length  $L_{\text{ref}} = 32768$ , with  $s = 2$  and fixed amplitude  $g = 1 + 0.074 \log 2 = 1.0512928914$ . The YaRN-2 comparison uses the Transformers official-equation table and  $g = 1 + 0.1 \log 2 = 1.0693147181$ ; Native uses  $g = 1$ .

The full RULER-13 confirmation uses seed 202609027, 20 examples per task and length, the same tokenizer and native chat template, and greedy task-budgeted generation. It contains 520 generations per arm across 32768 and 65536 tokens. Per-task scores are shown in Table 22. The macro averages the thirteen task means equally. Paired intervals use 10,000 bootstrap samples within task strata with seed 202609028. The 64K index-minus-YaRN difference is 6.0897 percentage points, with interval  $[2.7627, 9.5835]$ ; at 32K it is  $-0.0256$  points, with interval  $[-3.2630, 3.2179]$ .

Table 22: Qwen-0.5B RULER-13 per-task score (%), 20 examples per cell.

| Task              | 32K    |        |        | 64K    |        |        |
|-------------------|--------|--------|--------|--------|--------|--------|
|                   | Native | Index  | YaRN   | Native | Index  | YaRN   |
| Single-1          | 100.00 | 100.00 | 100.00 | 75.00  | 100.00 | 100.00 |
| Single-2          | 100.00 | 100.00 | 100.00 | 15.00  | 100.00 | 75.00  |
| Single-3          | 100.00 | 100.00 | 100.00 | 30.00  | 100.00 | 95.00  |
| MultiKey-1        | 90.00  | 95.00  | 90.00  | 50.00  | 90.00  | 65.00  |
| MultiKey-2        | 50.00  | 60.00  | 35.00  | 0.00   | 40.00  | 25.00  |
| MultiKey-3        | 5.00   | 0.00   | 10.00  | 0.00   | 5.00   | 5.00   |
| MultiValue        | 62.50  | 81.25  | 81.25  | 22.50  | 63.75  | 58.75  |
| MultiQuery        | 62.50  | 80.00  | 77.50  | 22.50  | 65.00  | 50.00  |
| Variable tracking | 48.00  | 42.00  | 46.00  | 3.00   | 28.00  | 36.00  |
| Common words      | 2.50   | 2.00   | 2.50   | 7.00   | 0.50   | 0.00   |
| Frequent words    | 46.67  | 31.67  | 35.00  | 31.67  | 46.67  | 40.00  |
| QA-1              | 20.00  | 10.00  | 30.00  | 5.00   | 15.00  | 20.00  |
| QA-2              | 25.00  | 25.00  | 20.00  | 25.00  | 15.00  | 20.00  |

## F.2 PLACEMENT ON ANOTHER FINITE PAIR GRID

The alternative direct-gap construction evaluates the same clipped profile using the target table’s local frequency spacing:

$$c_K = (1 - b^{-1/K})^{-1}, \quad \xi_k^{(K)} = \log \frac{L_{\text{ref}} \omega_k^N}{2\pi c_K}, \quad m_k^{\text{gap}} = \text{clip} \left( \frac{\xi_H - \xi_k^{(K)}}{\xi_H - \xi_L}, 0, 1 \right).$$

Both placements share the frozen  $\xi_H, \xi_L$  and amplitude coefficient 0.074. Within each comparison, endpoints, scale, amplitude, checkpoint, inputs, decoder, and scorer are matched. Table 23 reports two independent-input confirmations with 80 rows per task.

Qwen-0.5B uses the 32-pair,  $s = 2$  setting above and data seed 202609026, with four tasks at each of 32K and 64K. These are separate inputs from the subsequent RULER-13 confirmation. Gemma-1.1-2B-Instruct uses  $K = 128$ ,  $b = 10,000$ , a fixed operating reference of 4096,  $s = 4$ , and  $g = 1 + 0.074 \log 4 = 1.1025857827$ . Its confirmation uses data seed 202609028 at 16K. The task vector is NIAH single-1, multikey-2, multikey-3, and variable tracking. Gemma’s direct-gap

scores are 100/88.75/20/82.50%; normalized-index scores are 100/88.75/40/87.25%. Both tables were frozen before these inputs were generated, and the earlier selection rows are excluded. This confirmation compares the two placements; the Native/YaRN baseline panel uses a different, earlier input set.

Table 23: Independent-input placement comparisons, 80 examples per task and four tasks per panel. Intervals are for index minus direct-gap, in percentage points: 97.5% for each Qwen length (Bonferroni over two lengths), 95% for Gemma.

| Model     | Length | Direct-gap (%) | Index (%) | Difference interval |
|-----------|--------|----------------|-----------|---------------------|
| Qwen-0.5B | 32K    | 49.69          | 53.37     | $[-0.25, +7.75]$    |
| Qwen-0.5B | 64K    | 46.62          | 46.13     | $[-4.87, +3.81]$    |
| Gemma-2B  | 16K    | 72.81          | 79.00     | $[+2.81, +9.62]$    |

### F.3 BOUNDARY-MATCHED INTERMEDIATE EXPONENTS

Let  $l$  be the last native frequency pair making more than 32 turns over the reference window and  $h$  the first making less than one turn. Write  $N = h - l$  and  $q = \text{clip}(k - l, 0, N)$ . All profiles leave the fast band unchanged and divide the slow-band frequencies by  $s$ :

$$\omega'_k = \omega_k^N s^{-m_q}, \quad m_0 = 0, \quad m_N = 1. \quad (53)$$

With the same band boundaries, the three intermediate allocations are

$$m_q^{\text{Uni}} = \frac{q}{N}, \quad (54)$$

$$m_q^{\text{Pro}} = \frac{q(q+1)}{N(N+1)}, \quad (55)$$

$$m_q^{\text{BM}} = \frac{q(q+1)(3N+2-2q)}{N(N+1)(N+2)}. \quad (56)$$

The Uni and Pro rules follow the uniform and progressive mixed-radix constructions of Tian et al. (2026). BM smooths the radix increments  $\epsilon_q = m_q - m_{q-1}$  at both band boundaries.

To obtain BM, minimize  $\sum_{q=0}^N (\epsilon_{q+1} - \epsilon_q)^2$  with  $\epsilon_0 = \epsilon_{N+1} = 0$  and  $\sum_{q=1}^N \epsilon_q = 1$ . The discrete Euler equation gives

$$\epsilon_q = \frac{6q(N+1-q)}{N(N+1)(N+2)};$$

summing gives (56). The quadratic form is strictly convex with the endpoints fixed, so this solution is unique. Figure 9 shows the actual profile shapes for the tested transition widths.

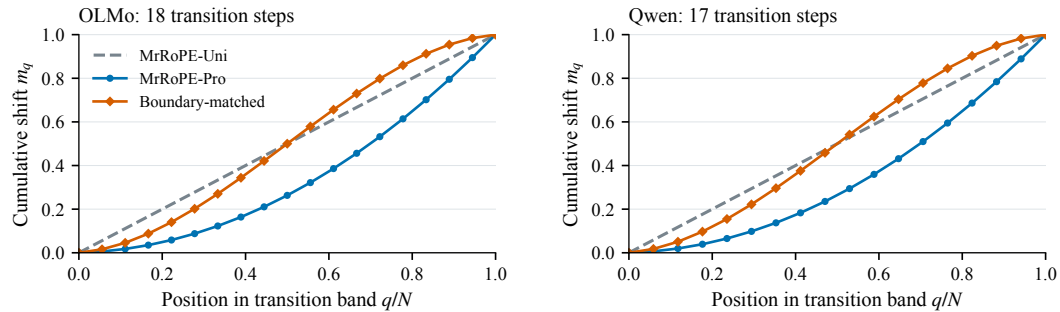


Figure 9: **Intermediate-band exponent displacements.** All three profiles have the same cumulative scale at the endpoints. BM redistributes the increments toward the middle of the transition band.

OLMo-2-0425-1B-Instruct uses  $b = 500,000$ , reference length 4096,  $K = 64$ , and  $(l, h, N) = (14, 32, 18)$ . Qwen-2.5-3B/7B-Instruct use  $b = 10^6$ , reference length 32768,  $K = 64$ , and  $(l, h, N) = (23, 40, 17)$ . The factor-four comparisons use  $g = 1 + 0.1 \log 4 = 1.1386294361$  for both MrRoPE-Pro and BM, with FP32 frequency tables and BF16 model execution.

#### F.4 SIX-TASK MODEL COMPARISONS

The shared task types are NIAH single-2, multikey-2, multiquery, variable tracking, frequent words, and QA-1. OLMo’s confirmation uses seed 20260910 with four short and eight long examples per task, after a separate development panel with seed 20260909. Its 4K/16K scores are listed in Table 4; MrRoPE-Uni on the same confirmation inputs scores 76.88/32.12%, and official YaRN scores 54.38/6.94%. The Qwen-3B panel has two 32K and four 128K examples per task. The 7B screen reuses the first one short and two long examples per task; this selection was fixed before its outputs were inspected. Table 24 reports every task mean.

All arms use the corresponding native chat template and the same task-specific generation caps, score the entire saved response with the RULER task metric, and retain generated token IDs and EOS flags. On Qwen-7B, a tokenwise MLP chunk size of 4096 makes 128K evaluation fit in memory; both arms share that path while retaining full attention.

Table 24: All six-task BM/MrRoPE-Pro comparisons. Values are official RULER task scores (%); model-specific sample counts are given in Table 4.

| Model   | Task              | Short  |        | Long   |        |
|---------|-------------------|--------|--------|--------|--------|
|         |                   | MrPro  | BM     | MrPro  | BM     |
| OLMo-1B | Single-2          | 75.00  | 100.00 | 0.00   | 100.00 |
|         | MultiKey-2        | 25.00  | 100.00 | 0.00   | 37.50  |
|         | MultiQuery        | 18.75  | 87.50  | 0.00   | 43.75  |
|         | Variable tracking | 25.00  | 70.00  | 0.00   | 10.00  |
|         | Frequent words    | 58.33  | 58.33  | 16.67  | 41.67  |
|         | QA-1              | 25.00  | 75.00  | 0.00   | 75.00  |
| Qwen-3B | Single-2          | 100.00 | 100.00 | 100.00 | 100.00 |
|         | MultiKey-2        | 100.00 | 100.00 | 75.00  | 50.00  |
|         | MultiQuery        | 100.00 | 100.00 | 93.75  | 100.00 |
|         | Variable tracking | 90.00  | 100.00 | 75.00  | 75.00  |
|         | Frequent words    | 83.33  | 100.00 | 75.00  | 75.00  |
|         | QA-1              | 50.00  | 50.00  | 50.00  | 25.00  |
| Qwen-7B | Single-2          | 100.00 | 100.00 | 100.00 | 100.00 |
|         | MultiKey-2        | 100.00 | 100.00 | 100.00 | 100.00 |
|         | MultiQuery        | 100.00 | 100.00 | 100.00 | 100.00 |
|         | Variable tracking | 100.00 | 80.00  | 90.00  | 60.00  |
|         | Frequent words    | 100.00 | 100.00 | 66.67  | 66.67  |
|         | QA-1              | 0.00   | 0.00   | 50.00  | 0.00   |

#### F.5 NATURAL QUESTION ANSWERING

The natural comparison reuses frozen, source-verified LongBench prompts (Bai et al., 2024). Inputs are untruncated, retain their source questions and references, and fit the 16K input-plus-generation budget. Selection follows the pre-existing source-hash order; no generated answer participates in selection. The complete eligible pool contains 778 questions, of which 631 have more than 4096 input tokens. Both methods retain the same factor-four table at every length.

Greedy generation uses a 32-token cap for HotpotQA and 2WikiMQA, 128 for Qasper and NarrativeQA, and 64 for MultiFieldQA-en. The metric is token F1 on the complete saved generated response against the task references. Table 25 gives the long-input and original-window-length strata, including EOS counts. The long-input macro averages the five task means; the shorter-input macro averages its four available tasks. The respective BM-minus-MrPro changes are +3.8194 and +2.0028 points, with paired within-task bootstrap intervals [1.3161, 6.2920] and [−3.5618, 7.4742]. The MultiFieldQA long-input row includes 40 capped responses in each arm; these remain in the F1 calculation.

Table 25: Natural QA under static  $s = 4$ . Token F1 uses the whole generated response; EOS counts are recorded separately.

| Input length | Task         | $n$ | MrPro F1 (%) | BM F1 (%) | MrPro EOS | BM EOS |
|--------------|--------------|-----|--------------|-----------|-----------|--------|
| 4K–16K       | HotpotQA     | 166 | 28.63        | 33.54     | 160       | 163    |
|              | 2WikiMQA     | 173 | 21.55        | 25.43     | 169       | 171    |
|              | Qasper       | 119 | 13.94        | 19.06     | 119       | 119    |
|              | NarrativeQA  | 61  | 9.51         | 11.48     | 61        | 61     |
|              | MultiFieldQA | 112 | 34.49        | 37.72     | 72        | 72     |
| $\leq 4K$    | HotpotQA     | 5   | 58.33        | 51.67     | 5         | 5      |
|              | 2WikiMQA     | 26  | 34.62        | 44.98     | 26        | 26     |
|              | Qasper       | 78  | 26.16        | 28.14     | 78        | 78     |
|              | MultiFieldQA | 38  | 50.94        | 53.27     | 35        | 33     |

## G SUPPORTING RESULTS

### G.1 MULTI-HEAD LATENT ATTENTION

We report allocation at one fixed rotary budget in a 432M-parameter transformer following the MLA architectural pattern (DeepSeek-AI, 2024a), with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has  $d_{\text{rope}}=32$  ( $K=16$  frequencies); the remaining  $d_{\text{nope}}=32$  dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens (Penedo et al., 2024) at  $L_{\text{train}}=8192$  and base 500K over seeds 42, 43, and 88. EVQ-COSH uses the pre-specified  $\tau = 1.414$  setting. At 16K, Geo PPL is 141.1/132.5/142.8 for seeds 42/43/88, whereas EVQ-COSH is 93.7/92.7/100.3; all three paired directions agree.

For the supporting overlay, let  $\lambda_k = 2\pi/\omega_k$  and  $t_k = \text{clip}((\lambda_k - L_{\text{train}})/(31L_{\text{train}}), 0, 1)$ . The MLA wavelength-blend operator sets  $\omega'_k = \omega_k/[1 + t_k(s - 1)]$ : it preserves wavelengths below the training length, applies the full factor  $s$  above  $32L_{\text{train}}$ , and interpolates between them.

Table 26: MLA validation (432M,  $d_{\text{rope}}=32$ , 3-seed mean  $\pm$  std). Rows marked “MLA wavelength-blend” use this run’s wavelength-blend operator defined above.

| Method                                                      | PPL@8K         | PPL@16K                        | PPL@24K                          | PPL@32K                          |
|-------------------------------------------------------------|----------------|--------------------------------|----------------------------------|----------------------------------|
| Geo                                                         | 35.4 $\pm$ 0.9 | 138.8 $\pm$ 5.5                | 241.5 $\pm$ 2.6                  | 323.7 $\pm$ 5.2                  |
| Geo + MLA wavelength-blend ( $s = 4$ )                      | 35.5 $\pm$ 0.9 | 117.9 $\pm$ 6.5                | 204.9 $\pm$ 6.9                  | 278.5 $\pm$ 6.3                  |
| <b>EVQ-COSH</b>                                             | 35.8 $\pm$ 0.8 | <b>95.6<math>\pm</math>4.1</b> | 204.9 $\pm$ 14.7                 | 291.6 $\pm$ 20.5                 |
| <b>EVQ-COSH + MLA wavelength-blend (<math>s = 4</math>)</b> | 35.8 $\pm$ 0.8 | <b>71.1<math>\pm</math>4.1</b> | <b>153.2<math>\pm</math>12.5</b> | <b>236.6<math>\pm</math>15.9</b> |

EVQ-COSH reduces 16K PPL by 31.1% (138.8  $\rightarrow$  95.6), while 8K PPL changes by +0.9%. All three training seeds favour EVQ-COSH at 16K. The EVQ-COSH plus MLA wavelength-blend row has the lowest mean PPL at 16K and beyond, showing that the allocation also composes with this wavelength-dependent extension.